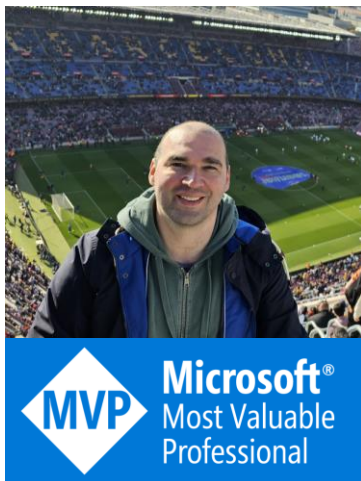


O'REILLY®

Microsoft Power BI Data Analyst Bootcamp (PL-300)

Learn in-demand skills and become a
certified Power BI Data Analyst





Nikola Ilic

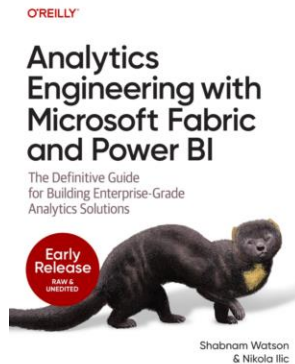
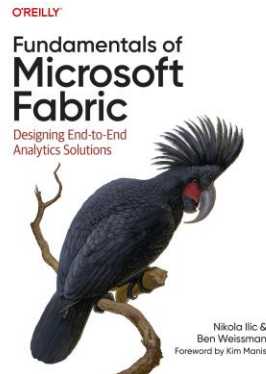
Consultant & Trainer



data-mozart.com

learn.data-mozart.com

- *I'm making music from the data!*
- Power BI and SQL addict, blogger, speaker...
- Father of 2, **Barca** & Leo Messi fan...





Schedule Day 1

- Introduction
- Prepare and model the data
 - Get data into Power BI
 - Data cleaning
 - Data transformation and loading
 - Design data model
 - Enhance data model with DAX
 - Optimize model performance
- Hands-on Lab



Schedule Day 2

- Visualize and Analyze the Data, Manage Workspaces, Secure Power BI Items
 - Create reports
 - Enhance reports for usability and storytelling
 - Identify patterns and trends in data
 - Create and manage workspaces and assets
 - Secure and govern Power BI items
 - Practical exam tips and quiz
- Hands-on Lab



Learning Objectives

- Provide actionable insights from the data
- Understand various components of Microsoft Power BI
- Enable data-driven decision making
- Get ready for the PL-300 exam (Microsoft Power BI Data Analyst)
- Practice and build your own Power BI solution



What's Your Experience With Microsoft Power BI? (POLL)

1. What is Power BI?
2. Heard some (cool) people talking about it, but I never tried it myself
3. Created a few reports
4. Can't start my day without coffee and Power BI
5. Super-duper Power BI user

The background image shows a person with glasses and a beanie sitting on a yellow bench, leaning over a yellow desk and working on a laptop. They are positioned in front of a large window that looks out onto green trees. The scene is overlaid with large, semi-transparent orange and yellow circles. The floor is a light-colored wood, and a small black stool is visible in the foreground.

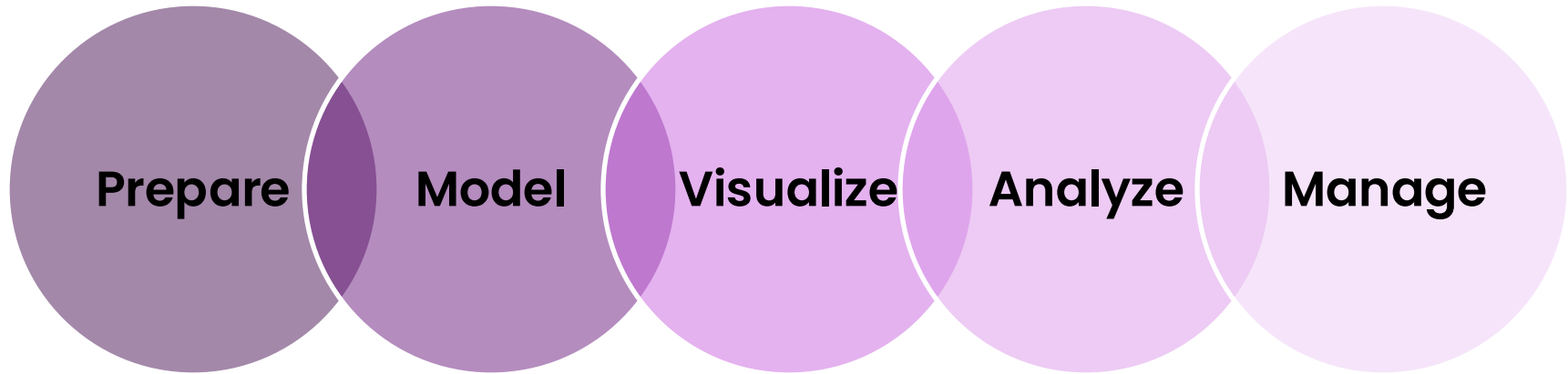
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Day 1

Microsoft Microsoft Power BI Data
Analyst Bootcamp (PL-300)



Tasks of a Data Analyst

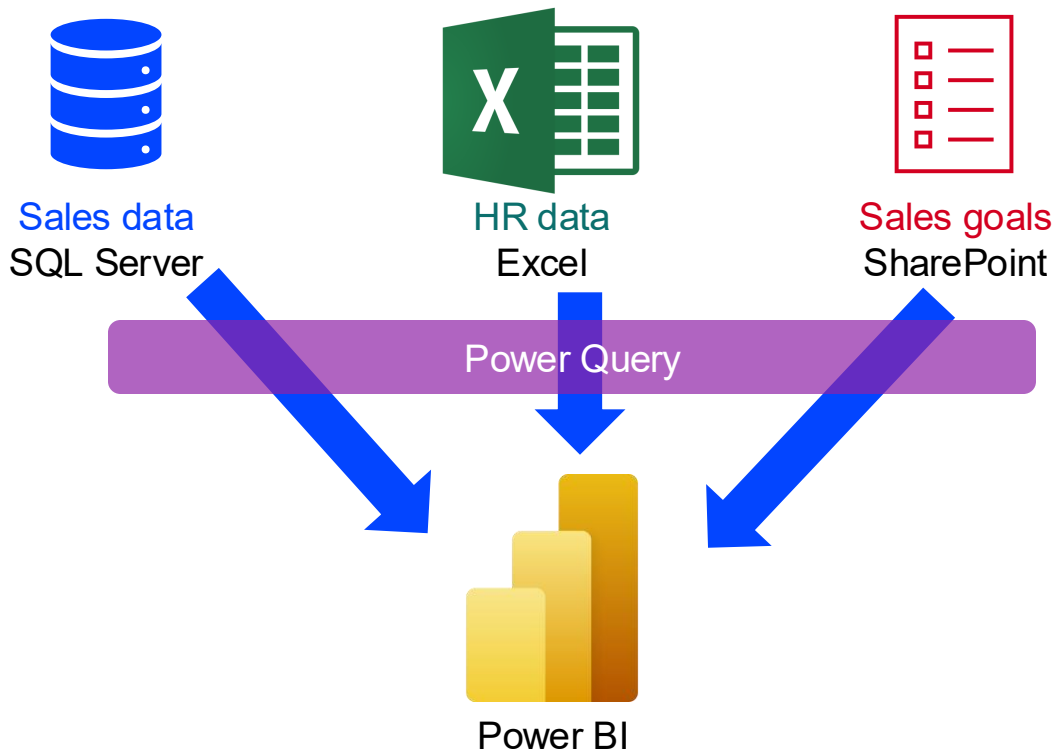




Identify and Connect to a Data Source



Combine Data from Multiple Sources





Shared vs. local semantic model



Local Semantic Model

- ✓ Created in the Power BI Desktop
- ✓ May include data from multiple disparate sources
- ✓ Once published to Power BI Service, it can be shared with others



Shared Semantic Model

- ✓ Stored in the Power BI Service
- ✓ Live connection from Power BI Desktop
- ✓ Enables separation of reports and semantic layer



Creating Reusable Power BI Assets



Product

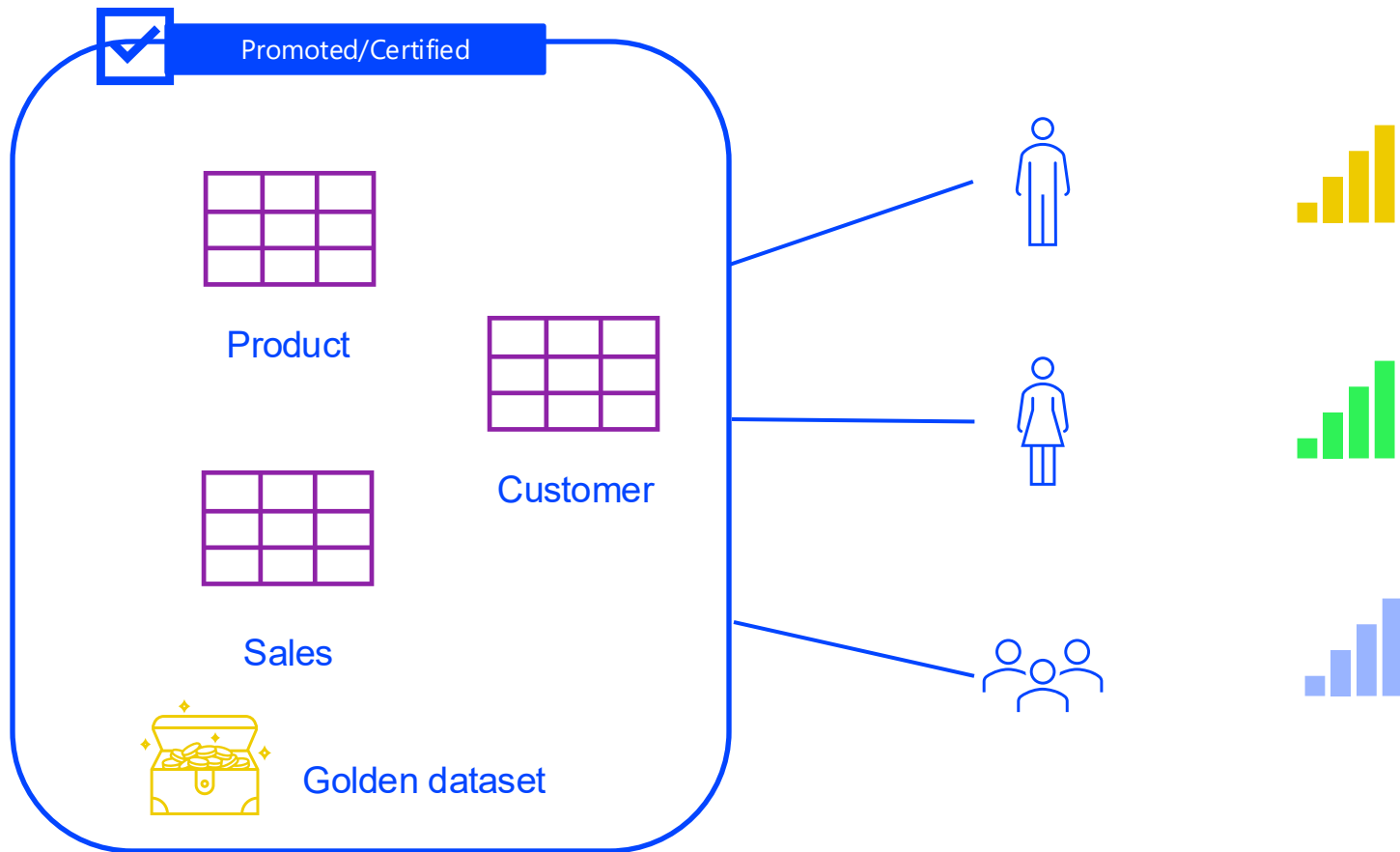


Sales



Customer

Creating Reusable Power BI Assets



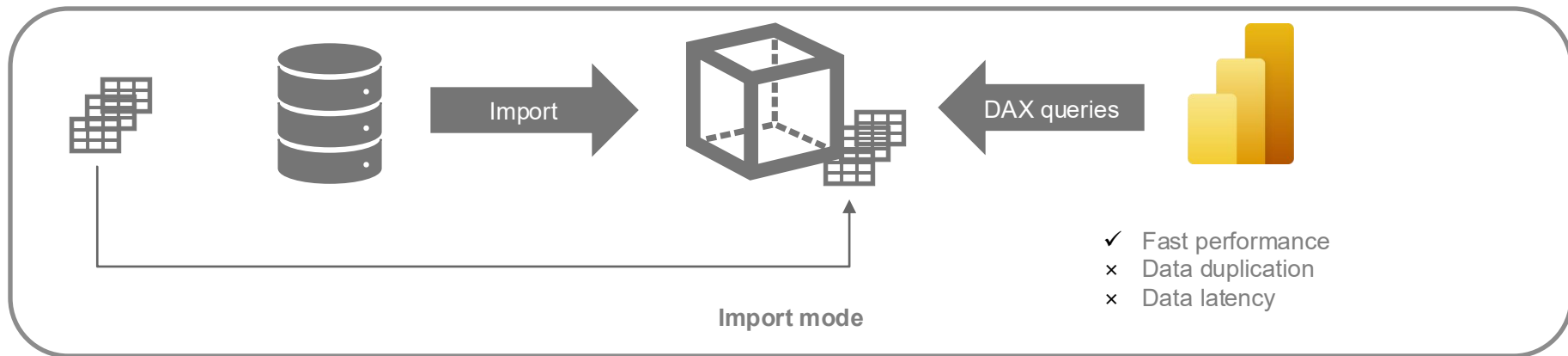


Understanding storage modes



Storage Modes – Import

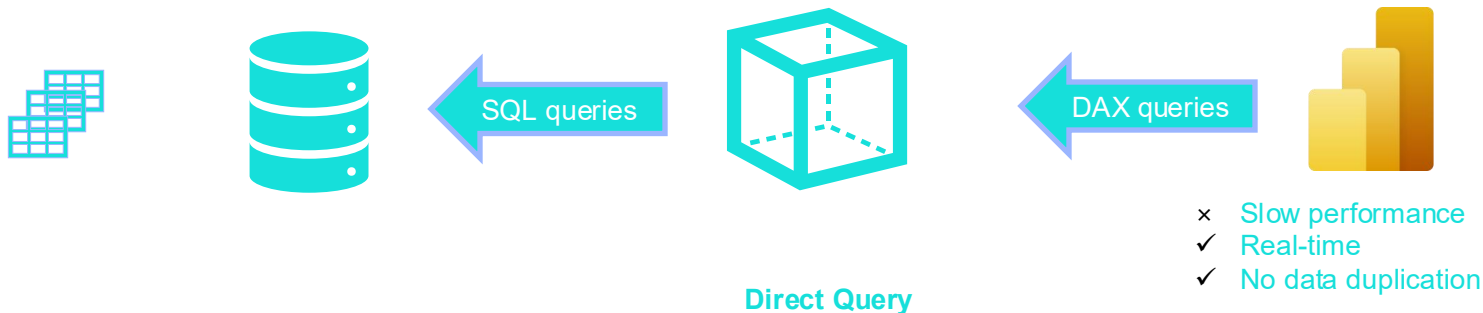
- ✓ Data stored in columnar in-memory database (VertiPaq)
- ✓ Snapshot at the specific point in time
- ✓ Periodically refreshed





Storage Modes – DirectQuery

- ✓ Data resides in the original source before, during, and after the query execution
- ✓ Data retrieved from the source *at the query time*





Storage Modes – Dual

- ✓ Combination of Import and DirectQuery
- ✓ Data is in-memory, but can also be retrieved directly from the source



Working with parameters



Working with Parameters

- ✓ Reusability – store once and use multiple times!
- ✓ Flexibility – dynamically change the query output

Transformation
step argument

Custom function
argument

Incremental
refresh



DEMO

- Connect to various data sources
- Change data source settings
- Connect to a shared model
- Set the storage mode





Q&A





Break



Profile and Clean Data

***“Systematic upfront analysis of
the data source content”***

Ralph Kimball



Data Profiling with Power Query

➤ Column quality (5)

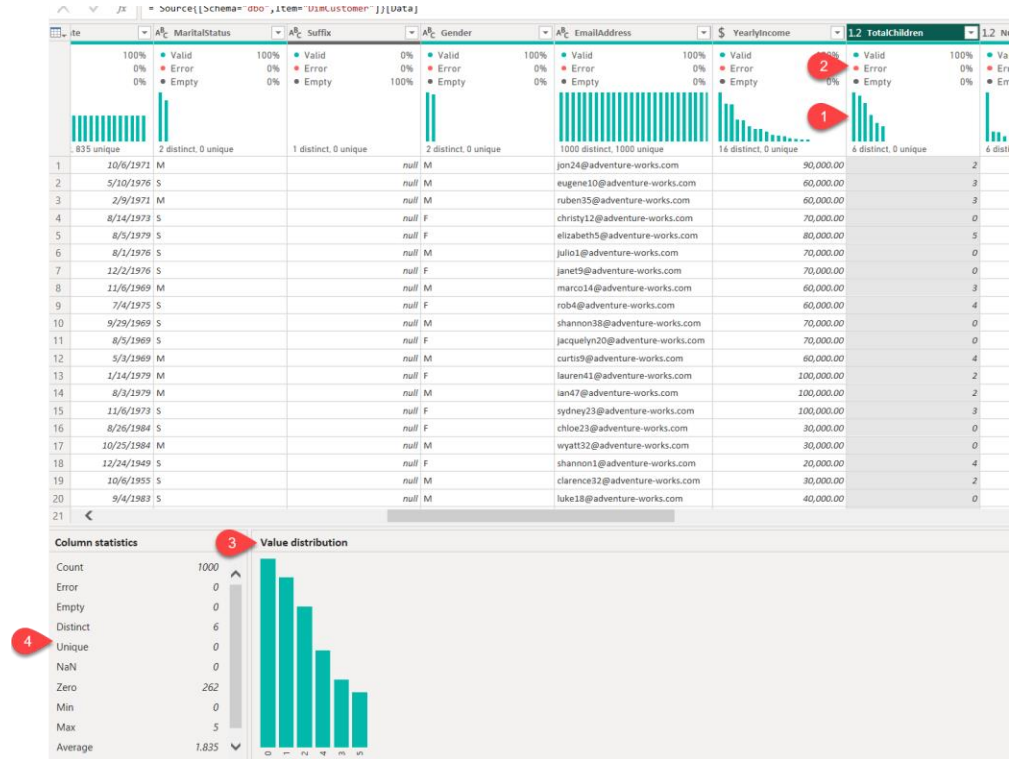
- Valid
- Error
- Empty

➤ Column distribution (5)

- Distinct vs Unique

➤ Column profile (3, 4)

- Value distribution
- Column statistics

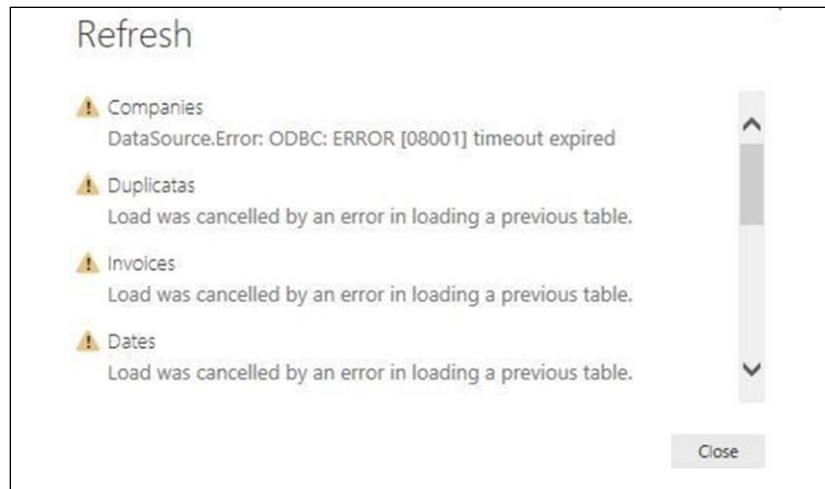




Data import errors

Possible data loading errors

- Query Timeout
- Couldn't find data formatted as a table
- Could not find file
- Data type errors





DEMO

- Data profiling with Power Query
- Resolve data inconsistencies
- Resolve data loading errors





Transform and Load Data



**Power Query is your
best data
transformation
friend!**





Data Types in Power BI

Power BI tries to convert to optimal data types

Number types

Decimal
(up to 15
digits)

47.0003852

Whole

47

Fixed
Decimal
(up to 4
digits)

47.0003

Date/time types

Date/Time

Date

Time

Date/Time
/Timezone

Duration



Data Types in Power BI

Text type

➤ ***Case sensitivity***

- Storage is case INsensitive -> A = a
- Power Query is case sensitive -> A <> a

➤ ***Leading and trailing spaces***

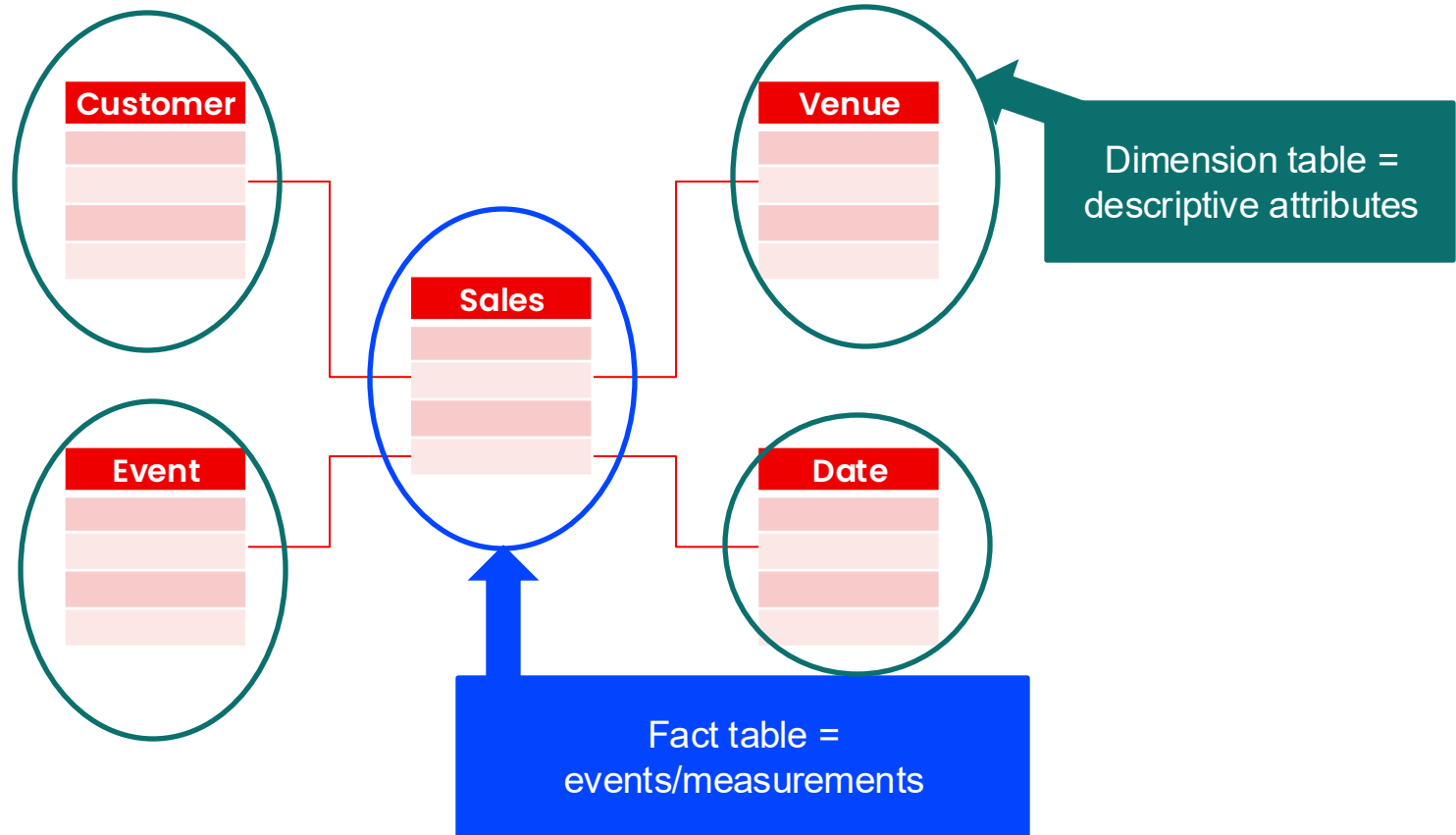
- Trailing spaces automatically trimmed
- Leading spaces aren't removed



Do you know what Star schema is? (POLL)

1. Star schema...What?!
2. I heard about it, but have no idea about how it's being used
3. I'm familiar with the concept and already implemented it
4. I'm a Star-schema "galactico"

Star Schema in a Nutshell





Why Star Schema?



- ✓ Easy navigation through the model
- ✓ Allows easier and faster data model modifications
- ✓ Consistent analysis
- ✓ Power BI engine is optimized for star schema models!



**Star schema should
always be a default
choice!**





Reference vs. Duplicate in Power Query

Duplicate

- ***Duplicate the query code***
 - Changes/add steps to duplicated query don't impact the base query
 - Original and duplicate co-exist as independent queries
 - Test changes while still maintaining a copy of original steps

Reference

- ***Refer the result of the query***
 - Original query remains as a base
 - Changes to the original query will impact the referenced query
 - Derive fact/dimension tables from the flat table



Append vs Merge

2023

Product	Date	Customer	Sales Amt

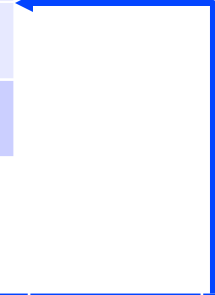


2024

Product	Date	Customer	Sales Amt

Product	Color	Name

Product table



Product	Category	Subcategory	Description	On Sale

Product Category table

Append vs Merge



Product	Date	Customer	Sales Amt

2023_2024

Append adds ROWS to the existing table

Product	Color	Name	Category	Subcategory	Description	On Sale

Product table - merged

Merge adds COLUMNS to the existing table



Merge Query Options

Table A

Fruit
Apple
Orange
Banana

Table B

Fruit
Apple
Orange
Mango

Left Outer

Fruit	Fruit
Apple	Apple
Orange	Orange
Banana	

Right Outer

Fruit	Fruit
Apple	Apple
Orange	Orange
	Mango

Inner

Fruit	Fruit
Apple	Apple
Orange	Orange

Full Outer

Fruit	Fruit
Apple	Apple
Orange	Orange
Banana	
	Mango



Understanding Column Keys



- ✓ Unique identifier for a table record
- ✓ Used for establishing relationships and performing JOIN operations



Disable Data Loading

- ✓ Disabled query/table will not be available in the report
- ✓ Useful for intermediate/staging tables



Understanding Query Folding

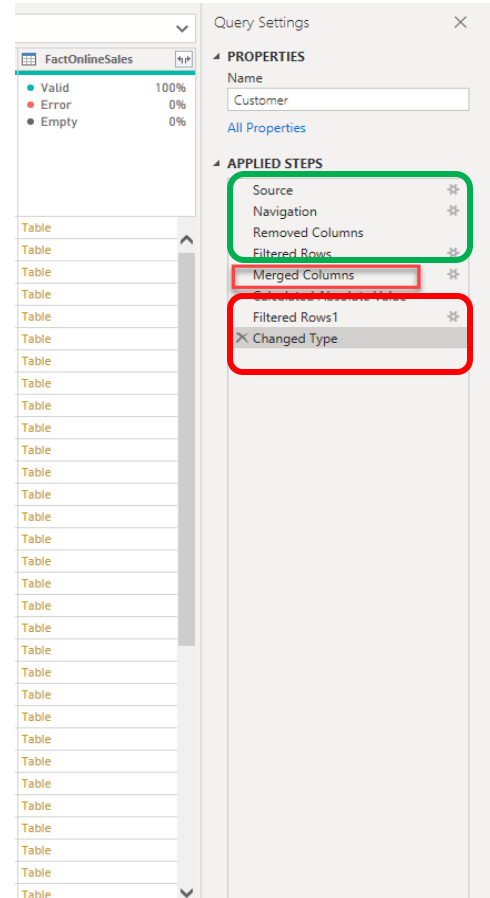
- ✓ Translate M to a query language
- ✓ Pushing heavy lifting on the data source
- ✓ *Ability to generate a single SQL query that's going to be executed on the data source side!*



Query Folding – Why Should I Care?

- ✓ Data refresh more efficient
- ✓ With DirectQuery/Dual the query must fold
- ✓ Incremental refresh serves its purpose

✓ ~~ALL or NOTHING?~~





Query Folding and Data Sources

- ✓ Relational databases, Odata, SSAS, SharePoint lists...
- ✓ Excel, text files...don't support query folding!
- ✓ *Not all data transformations will fold*

- ✓ Removing/renaming columns
- ✓ Filtering rows
- ✓ Grouping/Summarizing
- ✓ Merging/Appending queries from the same source
- ✓ Simple Custom columns
- ✓ Pivot/Unpivot



- ✓ Merging/Appending queries from different sources
- ✓ Complex Custom columns
- ✓ Index columns
- ✓ Changing column data type ("it depends")!





DEMO

- Common data transformations with Power Query
- Combining queries
- Managing parameters
- Query folding
- Implement a star schema with Power Query





Q&A





Break



Design Data Model



Relationships serve as filters





Relationship Cardinality

One-to-many (1:*)

Many-to-one (*:1)

One-to-one (1:1)

Many-to-many (*:*)

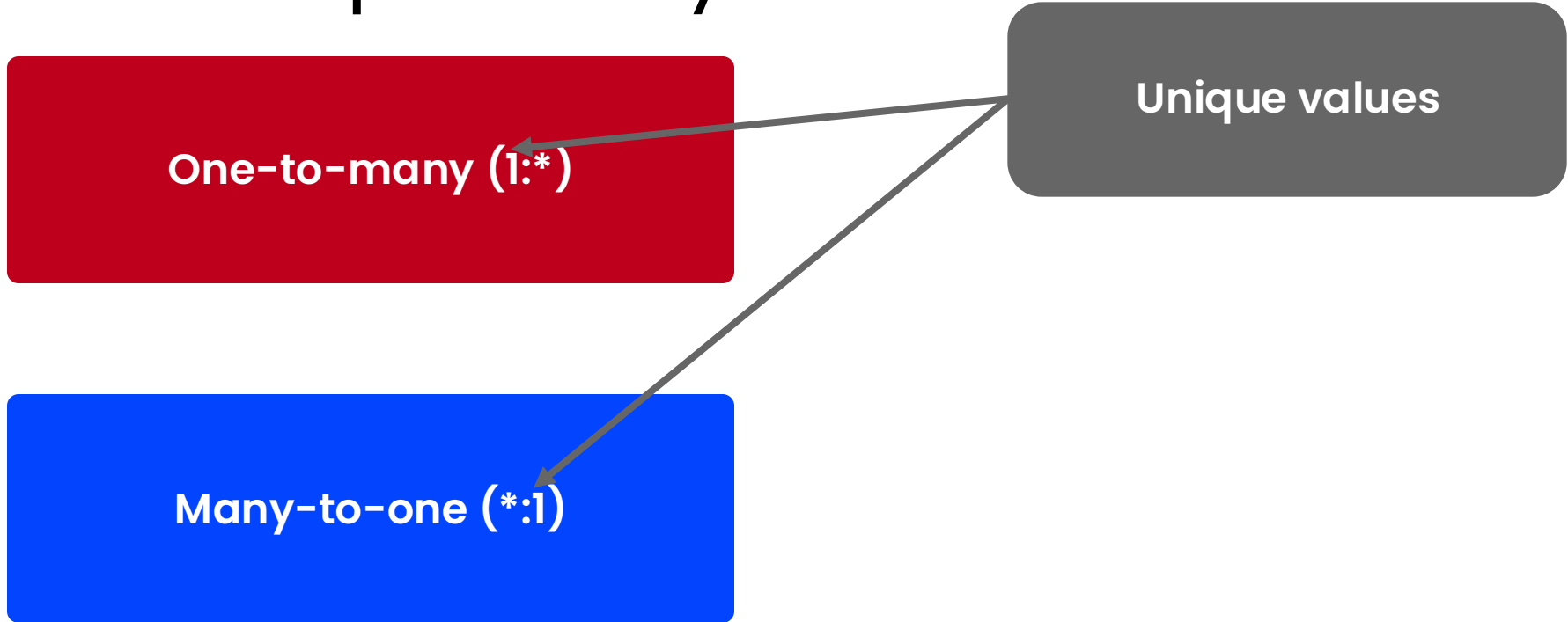


Relationship Cardinality

One-to-many (1:*)

Unique values

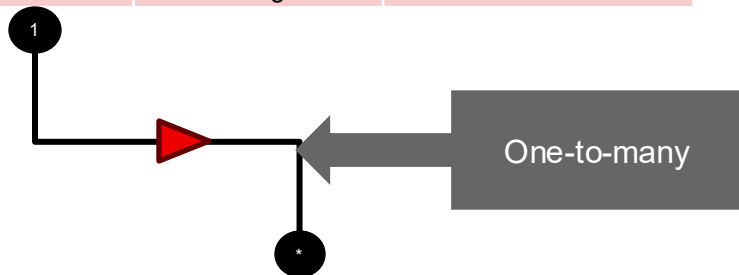
Many-to-one (*:1)





Relationship Cardinality (1:*, *:1)

Product Key	Product Name	Product Category
1	T-Shirt	Clothes
2	Socks	Clothes
3	Mug	Accessories

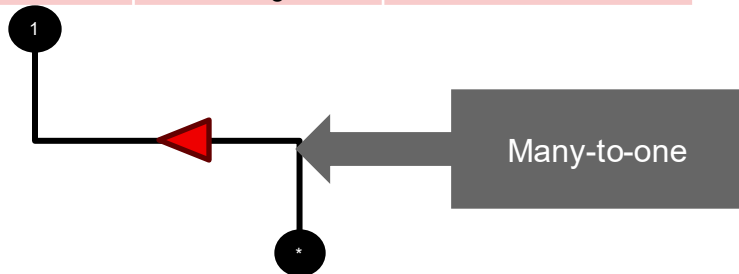


Sales Key	Product Key	Sales Amount
1	1	100
2	1	200
3	1	300
4	2	50
5	2	75
6	3	100
7	3	150



Relationship Cardinality (1:*, *:1)

Product Key	Product Name	Product Category
1	T-Shirt	Clothes
2	Socks	Clothes
3	Mug	Accessories

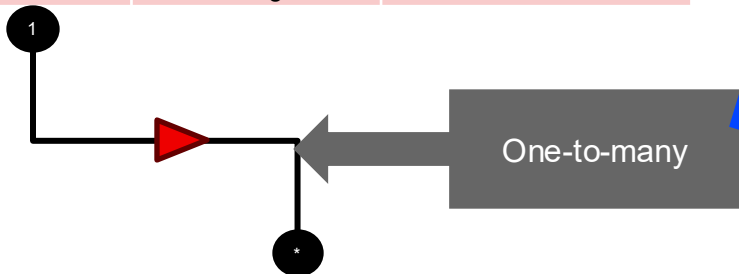


Sales Key	Product Key	Sales Amount
1	1	100
2	1	200
3	1	300
4	2	50
5	2	75
6	3	100
7	3	150



Relationship Cardinality (1:*, *:1)

Product Key	Product Name	Product Category
1	T-Shirt	Clothes
2	Socks	Clothes
3	Mug	Accessories

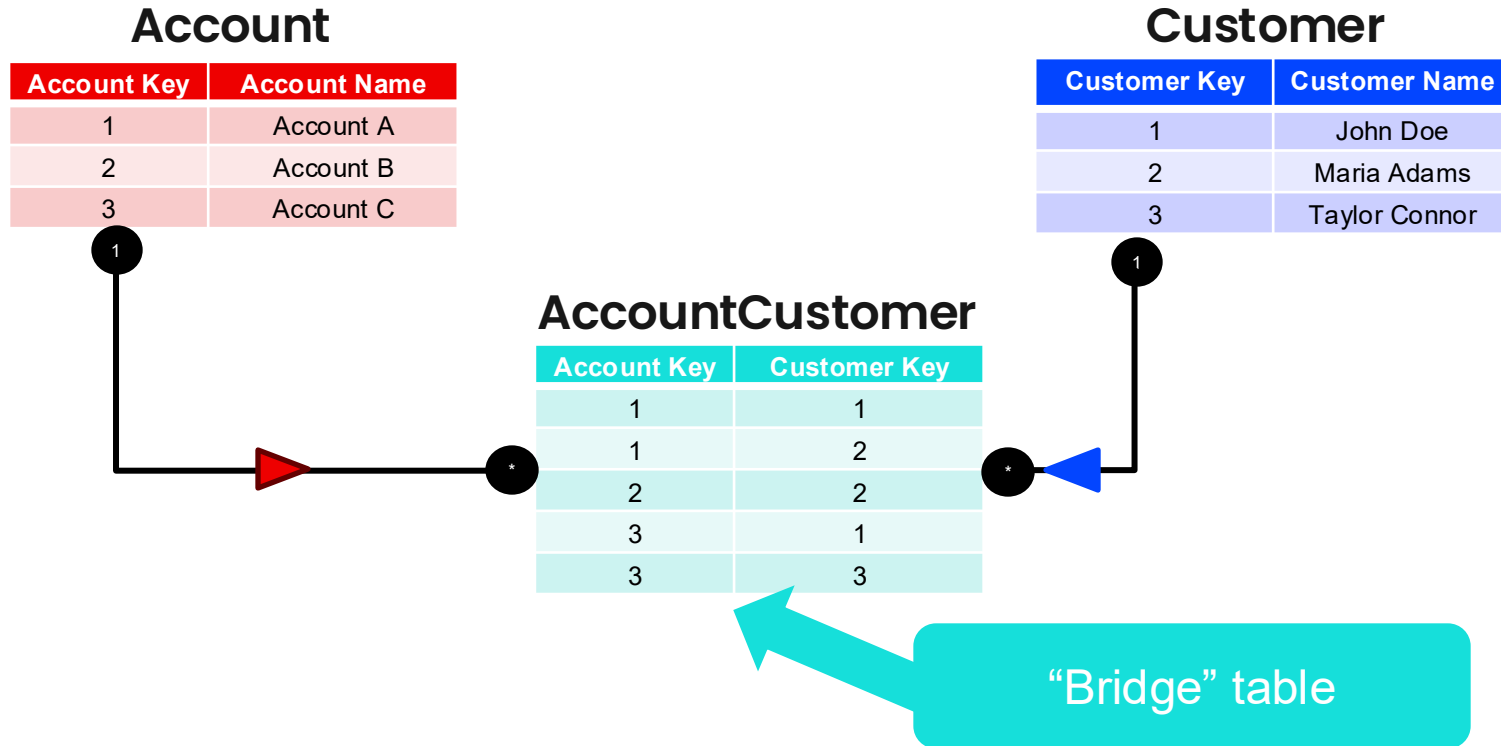


Preferred choice in Power BI!

Sales Key	Product Key	Sales Amount
1	1	100
2	1	200
3	1	300
4	2	50
5	2	75
6	3	100
7	3	150



Relationship Cardinality (*:*)



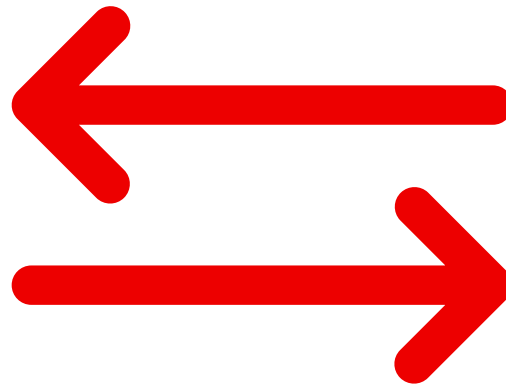


Relationship Filter Direction



Single Direction

Recommended practice



Both Directions

Implement with care



Relationship Filter Direction (Single)

Product Category Key	Product Category
1	Clothes
2	Accessories

1



*

Product Key	Product Name	Product Category Key
1	T-Shirt	1
2	Socks	1
3	Mug	2

1

Sales Key	Product Key	Sales Amount
1	1	100
2	1	200
3	1	300
4	2	50
5	2	75
6	3	100
7	3	150

*



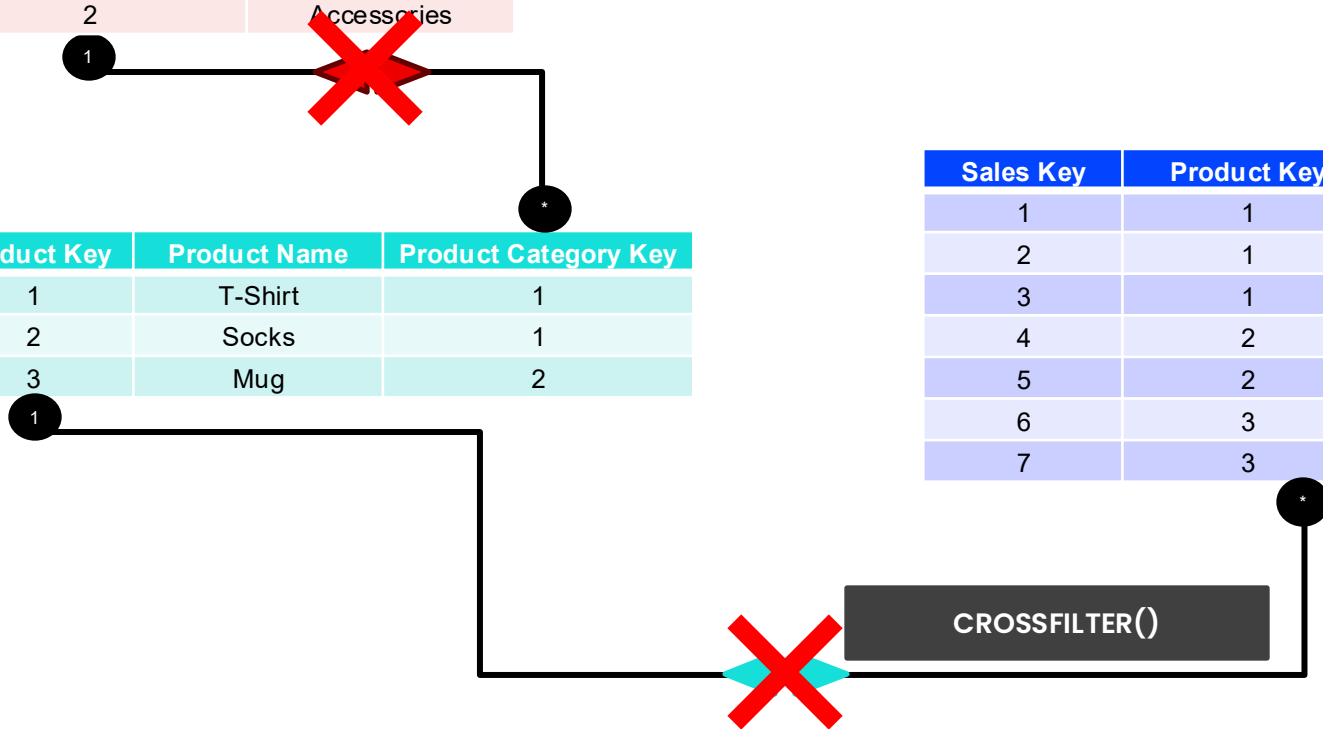
Relationship Filter Direction (Both)



Product Category Key	Product Category
1	Clothes
2	Accessories

Product Key	Product Name	Product Category Key
1	T-Shirt	1
2	Socks	1
3	Mug	2

Sales Key	Product Key	Sales Amount
1	1	100
2	1	200
3	1	300
4	2	50
5	2	75
6	3	100
7	3	150





CROSSFILTER() DAX Function

- ✓ Override default filter direction
- ✓ Valid only during the query execution!
- `CROSSFILTER(column1, column2, direction)`

Direction -> BOTH, NONE, ONEWAY*

**OneWay_LeftFiltersRight or OneWay_RightFiltersLeft*



Implement role-playing dimensions



Role-playing Dimension



- ✓ Dimension table “plays” multiple roles in the model
- ✓ Commonly used for date-based scenarios
 - Order date
 - Ship date
 - Due date



USERELATIONSHIP() DAX Function

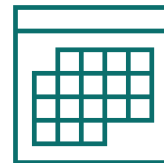
- ✓ Activate inactive relationship
- ✓ Can't be used when RLS is defined on the table!
- `USERELATIONSHIP(column1, column2)`



Create a common date table



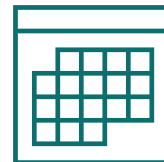
Date Table in Semantic Model



- ✓ Analyze data by date attributes (calendar year, fiscal year, quarter, month, week...)
- ✓ Necessary for applying DAX time-intelligence functions



How to Create a Date Table?



- ✓ Source system – data warehouse
- ✓ Power Query
- ✓ DAX
 - CALENDAR
 - CALENDARAUTO



DEMO

- Implement a role-playing dimension
- Configure relationship cardinality and cross-filter direction
- Create a calendar table





Enhance Data Model With DAX



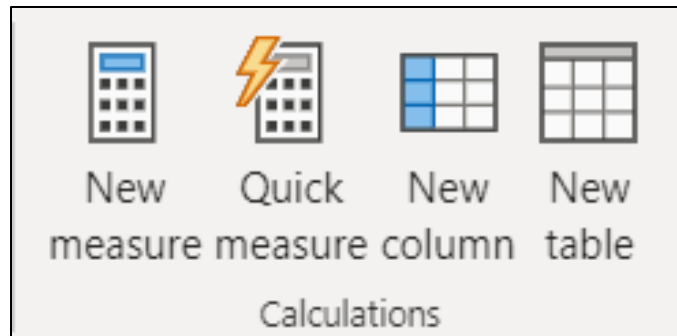
What's Your Experience With DAX? (POLL)

1. What is DAX?
2. I heard of DAX, but never used it
3. I'm familiar with basic DAX concepts
4. I know that I should “filter columns and not tables” in DAX
5. “I speak DAX like an Italian” 😊



DAX Essentials

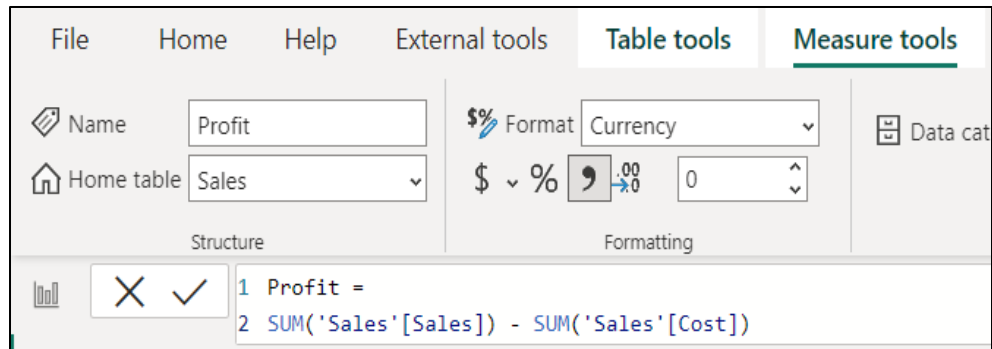
- Data Anal^ysis Expressions
- Build formulas and expressions with functions and operators
- Create measures, calculated columns, and calculated tables





Creating DAX measures

- Defined with DAX definitions
- Computed “on the fly”
- Not stored in the data model
- Responsive to interactions
- Indicated by calculator icon



-  Total Collisions
-  Total Injured
-  Total Killed



Implicit vs Explicit vs Quick measures

☐		Sales % All Region
☐		Sales % Country
☐		Sales % Group
☐		Sales
☐		ShipDate

Quick measure

Calculation

Filtered value 

Calculate a value with a filter applied. [Learn more](#)

Base value ⓘ

Add data fields here

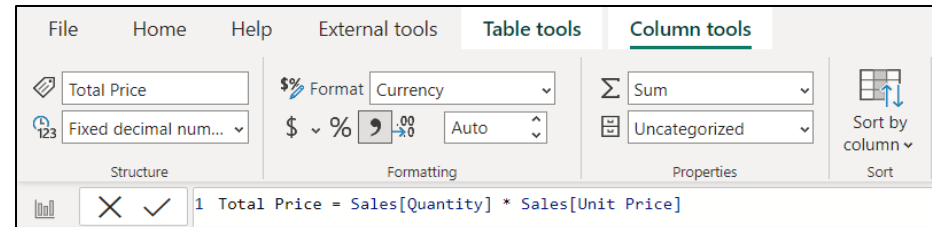
Filter ⓘ

Add data fields here



Creating Calculated Columns

- Defined using DAX expressions
- Computed & stored in data model
- Useful for “helper/connector columns”
- Recalculated during data refresh
- Table and Sigma icon



Total Price
\$57.68
\$4,049.98
\$4,049.98
\$57.68
\$1,637.4
\$40.38
\$4,049.98



Creating Calculated Tables

- Defined using DAX expressions
- Computed & stored in data model
- Useful for aggregating data or creating custom tables
- Table and calculator icon

The screenshot shows the Microsoft Power BI Desktop interface. The 'Table tools' ribbon is active, displaying options like 'Mark as date table', 'Manage relationships', and 'New measure'. The 'Structure' pane on the left shows a table named 'Date' with columns: Date, Year, Quarter, Month, and MonthKey. The 'Data' pane on the right shows a search bar and a list of tables, including 'Date', 'Fiscal', 'Month', 'MonthKey', 'Quarter', and 'Year'. The 'Date' table is selected, and its data is displayed in the main view.

Date	Year	Quarter	Month	MonthKey
7/1/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/2/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/3/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/4/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/5/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/6/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/7/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/8/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/9/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/10/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707



Common DAX Functions for Calculated Tables

- ✓ CROSSJOIN
- ✓ UNION
- ✓ NATURALINNERJOIN/NATURALLEFTOUTERJOIN
- ✓ INTERSECT
- ✓ DISTINCT
- ✓ VALUES
- ✓ CALENDAR/CALENDARAUTO



Columns vs Measures in DAX

Calculated columns

- ✓ Create values for each row in table
- ✓ Store values in the .pbix file
- ✓ Increases data model size
- ✓ Row-by-row calculation can impact performance
- ✓ Must be referenced with measures for reuse

Measures

- ✓ Calculated on demand
- ✓ Calculated based on filters
- ✓ Doesn't affect data model size
- ✓ DAX expressions may still be suboptimal
- ✓ Can reference other measures directly for reuse



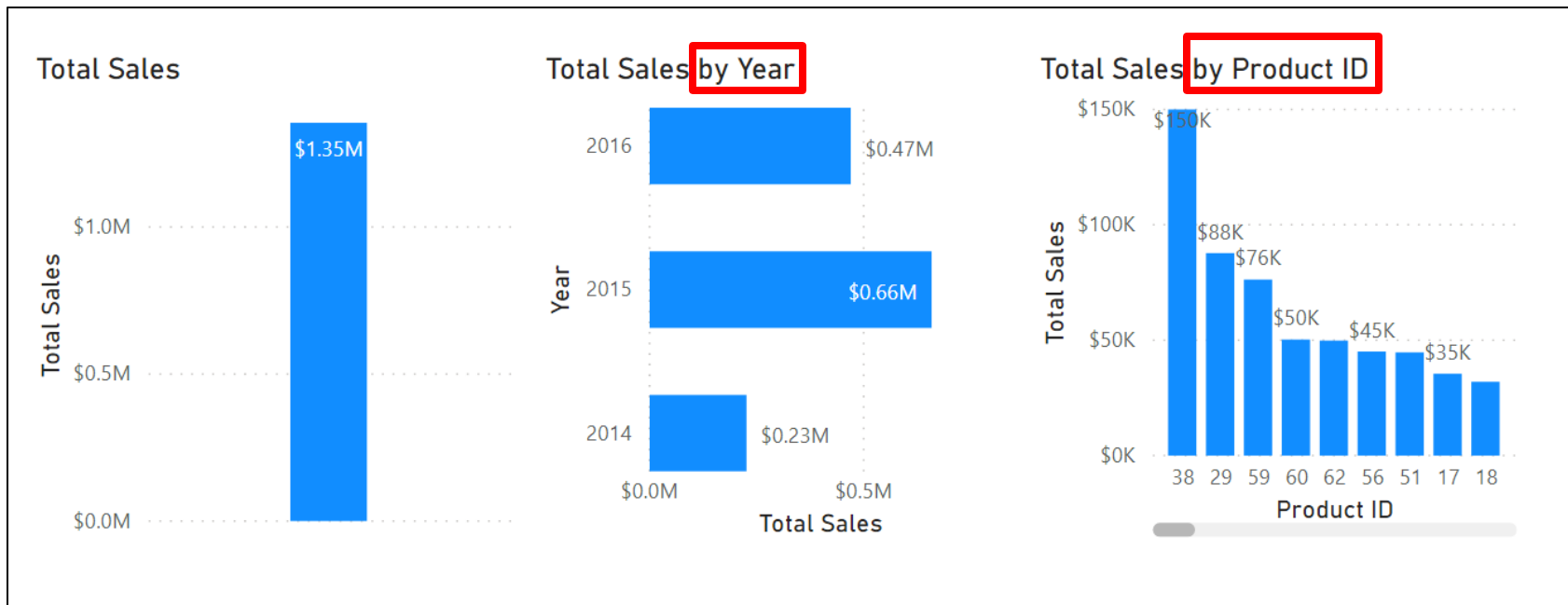
**CALCULATE is the only
function that can
change the filter
context!**





Understanding Filter Context

Measures are contextually different, or “dynamic,” depending on filters





Inside the CALCULATE Function

Existing filter context

```
2 CALCULATE(  
3     SUMX(  
4         Sales,  
5         Sales[SalesAmount]  
6     ),  
7     FILTER(  
8         ALL (Sales[ProductKey]),  
9         Sales[ProductKey] = 226  
10    ),  
11    FILTER(  
12        ALL(Sales[CustomerKey]),  
13        Sales[CustomerKey] = 123  
14    )  
15 )
```

Existing filter context

```
2 CALCULATE(  
3     SUMX(  
4         Sales,  
5         Sales[SalesAmount]  
6     ),  
7     ProductKey  
8         226  
9     CustomerKey  
10        123  
11 )
```

Modified filter context

ProductKey
226
CustomerKey
123

```
2 CALCULATE(  
3     SUMX(  
4         Sales,  
5         Sales[SalesAmount]  
6     ),  
7  
8  
9  
10  
11  
12  
13  
14  
15 )
```

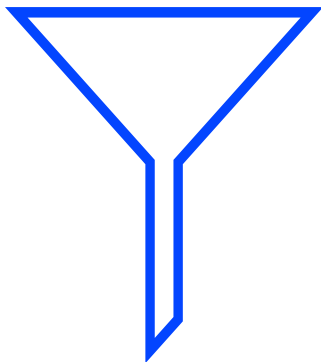
Modified filter context

ProductKey
226
CustomerKey
123

```
2 CALCULATE(  
3     SUMX(  
4         Sales,  
5         Sales[SalesAmount]  
6     ),  
7  
8  
9  
10  
11  
12  
13  
14  
15 )
```



Filter Modifiers Inside CALCULATE

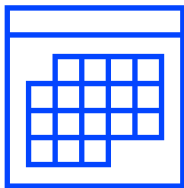


- **KEEPFILTERS**
- **USERELATIONSHIP**
- **CROSSFILTER**
- **ALL family of functions**
 - **ALL**
 - **ALLEXCEPT**
 - **ALLSELECTED**
 - **ALLNOBLANKROW**
 - **REMOVEFILTERS**



Prerequisites

Well-designed calendar table



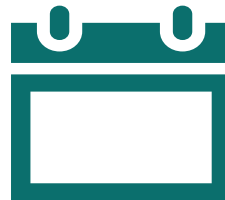
Continuous dates

Start on Jan 1st and end on Dec 31st including all dates in range



Date/Time or Date column

Unique values in the column



Mark as date table

If the relationship is not based on the date

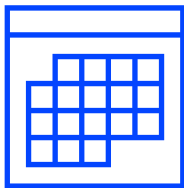


Implement Time-Intelligence Measures



Prerequisites

Well-designed calendar table



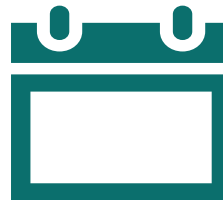
Continuous dates

Start on Jan 1st and end on Dec 31st including all dates in range



Date/Time or Date column

Unique values in the column



Mark as date table

If the relationship is not based on the date



Work with Semi-additive Measures



Semi-additive Measures in a Nutshell

- ✓ Can be summed across SOME dimensions (not all)
- ✓ Balance amounts – additive across all dimensions except time
- ✓ Bank balance and Inventory counts



Common DAX Functions for Semi-additive calculations

- ✓ FIRSTDATE/LASTDATE
- ✓ FIRSTNONBLANK/LASTNONBLANK
- ✓ OPENINGBALANCEYEAR/CLOSINGBALANCEYEAR/OPENINGBALANCEMONTH/
CLOSINGBALANCEMONTH



Create Calculation Groups

Calculation Groups in a Nutshell

Explicit measures

- ✓ Sales Amount
- ✓ Sales Quantity
- ✓ Discount Amount
- ✓ Return Quantity
- ✓

Business Logic

- ✓ Sales Amount in the previous quarter
- ✓ Sales Quantity YoY
- ✓ Running total of Return Quantity
- ✓

50 measures!



How Calculation Groups Work?

SELECTEDMEASURE()

Reference a measure that is currently in context: Sales Amount, Total Quantity...

YTD = CALCULATE(
SELECTEDMEASURE(),
DATESYTD(Date)
)

MTD = CALCULATE(
SELECTEDMEASURE(),
DATESMTD(Date)
)

Time Calculation CalendarYear	Current		YOY		YOY%	
	Sales	Orders	Sales	Orders	Sales	Orders
2012						
January	\$495,364	252	\$25,540	108	5.44%	75.00%
February	\$506,994	260	\$40,659	116	8.72%	80.56%
March	\$373,483	212	(\$111,716)	62	-23.02%	41.33%
April	\$400,336	219	(\$101,738)	62	-20.26%	39.49%
May	\$358,878	207	(\$202,804)	33	-36.11%	18.97%
June	\$555,160	318	(\$182,680)	88	-24.76%	38.26%
July	\$444,558	246	(\$152,188)	58	-25.50%	30.85%
August	\$523,917	294	(\$90,641)	101	-14.75%	52.33%
September	\$486,177	269	(\$116,906)	84	-19.38%	45.41%
October	\$535,159	313	(\$173,049)	92	-24.43%	41.63%
November	\$537,956	324	(\$122,590)	116	-18.56%	55.77%
December	\$624,502	487	(\$44,929)	265	-6.71%	119.37%
2013						
January	\$857,690	1662	\$362,326	1410	73.14%	559.52%
February	\$771,349	3453	\$264,355	3193	52.14%	1228.08%
March	\$1,049,907	4087	\$676,424	3875	181.11%	1827.83%
April	\$1,046,023	3979	\$645,687	3760	161.29%	1716.89%
May	\$1,284,593	4399	\$925,715	4192	257.95%	2025.12%
June	\$1,643,178	5025	\$1,088,018	4707	195.98%	1480.19%
July	\$1,371,676	4671	\$927,118	4425	208.55%	1798.78%
August	\$1,551,066	4865	\$1,027,148	4571	196.05%	1554.76%
September	\$1,447,496	4616	\$961,318	4347	197.73%	1615.99%
October	\$1,673,293	5300	\$1,138,134	4987	212.67%	1593.29%
November	\$1,780,920	5224	\$1,242,965	4900	231.05%	1512.35%
December	\$1,874,360	6231	\$1,249,858	5744	200.14%	1179.47%



DEMO

- Create calculated columns and measures
- Leverage the CALCULATE function to manipulate the filter context
- Implement time-intelligence DAX functions
- Create calculation groups





Q&A





Break

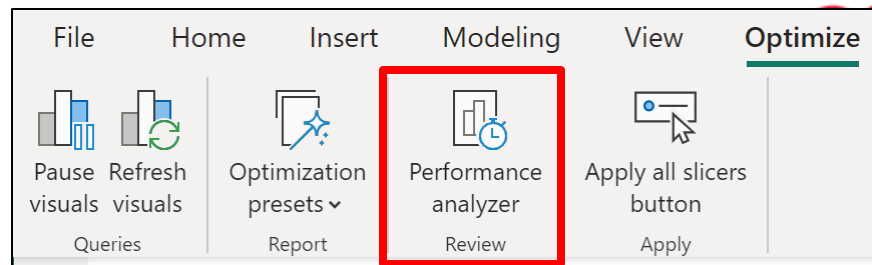


Optimize Model Performance

Performance Analyzer

Built-in tool in Power BI Desktop

- Record report interactions
- Review query and rendering times
- Identify bottlenecks and optimize queries



Performance analyzer

Start recording Refresh visuals Stop

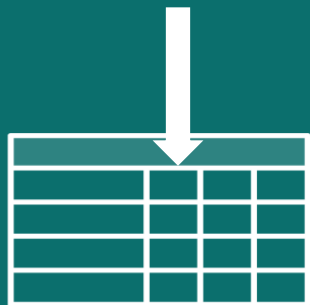
Clear Export

Name	Duration (ms)
Recording started (6/19/2023 3:12:20 PM)	-
Cross-highlighted	-
Button	92
Image	91
Sales and Profit Margin by Month	359
DAX query	19
Visual display	28
Other	312
Copy query	
Quantity by Category	302
Slicer	129
Slicer	222



Vertical vs Horizontal Filtering

Vertical

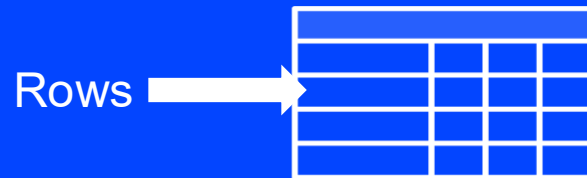


Columns

Model structure

Reporting design

Horizontal



Report consumer

Time dimension

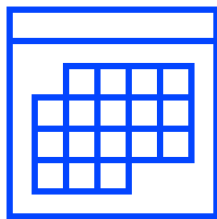
Filter by entity



Granularity
=
Level of detail in the dataset



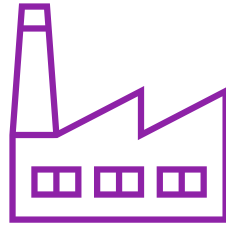
What Is the Lowest Acceptable Granularity?



Day



What Is the Lowest Acceptable Granularity?



Manufacturer



Changing Column Data Type: Dates

Purchase Date

2024-10-10 10:25

2024-10-10 17:45

2024-10-10 09:36

2024-10-10 11:44

2024-10-10 19:52

Cardinality = 5

Purchase Date

2024-10-10

2024-10-10

2024-10-10

2024-10-10

2024-10-10

Cardinality = 1



Changing Column Data Type: Decimal Numbers

Sales Amount

123.78612
123.78615
123.78618
123.78621
123.78624

Cardinality = 5

Sales Amount

123.78
123.78
123.78
123.78
123.78

Cardinality = 1



What's the largest table you have to deal with in Microsoft Power BI? (POLL)

1. < 10k records
2. < 1 million records
3. < 100 million records
4. >= 100 million records



Large Fact Table

Date	Customer ID	Product ID	Sales Amount
2024-10-12	123	11	10
2024-10-12	456	12	20
2024-10-12	789	12	50
2024-10-13	123	13	30
2024-10-13	456	11	10

100.000.000.....1.000.000.000 rows



Create Aggregated Tables

Date	Sales Amount
2024-10-12	80
2024-10-13	40

Product ID	Sales Amount
11	20
12	70
13	30

Customer ID	Sales Amount
123	40
456	30
789	50

Date	Customer ID	Product ID	Sales Amount
2024-10-12	123	11	10
2024-10-12	456	12	20
2024-10-12	789	12	50
2024-10-13	123	13	30
2024-10-13	456	11	10



Create Aggregated Tables

Date	Sales Amount
2024-10-12	80
2024-10-13	40

Product ID	Sales Amount
11	20
12	70
13	30

Customer ID	Sales Amount
123	40
456	30
789	50

Date	Customer ID	Product ID	Sales Amount
2022-10-12	123	11	10
2022-10-12	456	12	20
2022-10-12	789	12	50
2022-10-13	123	13	30
2022-10-13	456	11	10



Create Aggregated Tables

Date	Sales Amount
2024-10-12	80
2024-10-13	40

Product ID	Sales Amount
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Create Aggregated Tables

Date	Sales Amount
2024-10-12	80
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2022-10-13	456	11	10



Create Aggregated Tables

Date	Sales Amount
2024-10-12	80
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Product ID	Sales Amount
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2024-10-12	123	11	10
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2024-10-12	789	12	50
2024-10-13	123	13	30
2024-10-13	456	11	10



Create Aggregated Tables

Date	Sales Amount
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Product ID	Sales Amount
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Customer ID	Sales Amount
123	40
456	30
789	50

Date	Customer ID	Product ID	Sales Amount
2022-10-12	123	11	10
2022-10-12	456	12	20
2022-10-12	789	12	50
2022-10-13	123	13	30
2022-10-13	456	11	10



Reduce the number of rows



Reduce the model size



Aggregation “Awareness” in Power BI

Make Power BI “aware” of aggregated tables

Date	Sales Amount
2021-10-12	80
2021-10-13	40

Product ID	Sales Amount
11	20
12	70
13	30

Customer ID	Sales Amount
123	40
456	30
789	50



DEMO

- Use Performance Analyzer to identify problematic elements
- Optimize data model by changing data types





Q&A



It's your turn...



[Lab 1 Instructions](#)

[Lab 2 Instructions](#)

Hands-on Lab Part 1



The background image shows a person with glasses and a beanie sitting on a yellow bench, leaning over a yellow desk and working on a laptop. The scene is brightly lit by large windows showing green trees outside. The image is overlaid with large, semi-transparent orange and yellow circles. The text 'O'REILLY' is in white on the left side.

O'REILLY®

Day 2

Microsoft Microsoft Power BI Data
Analyst Bootcamp (PL-300)



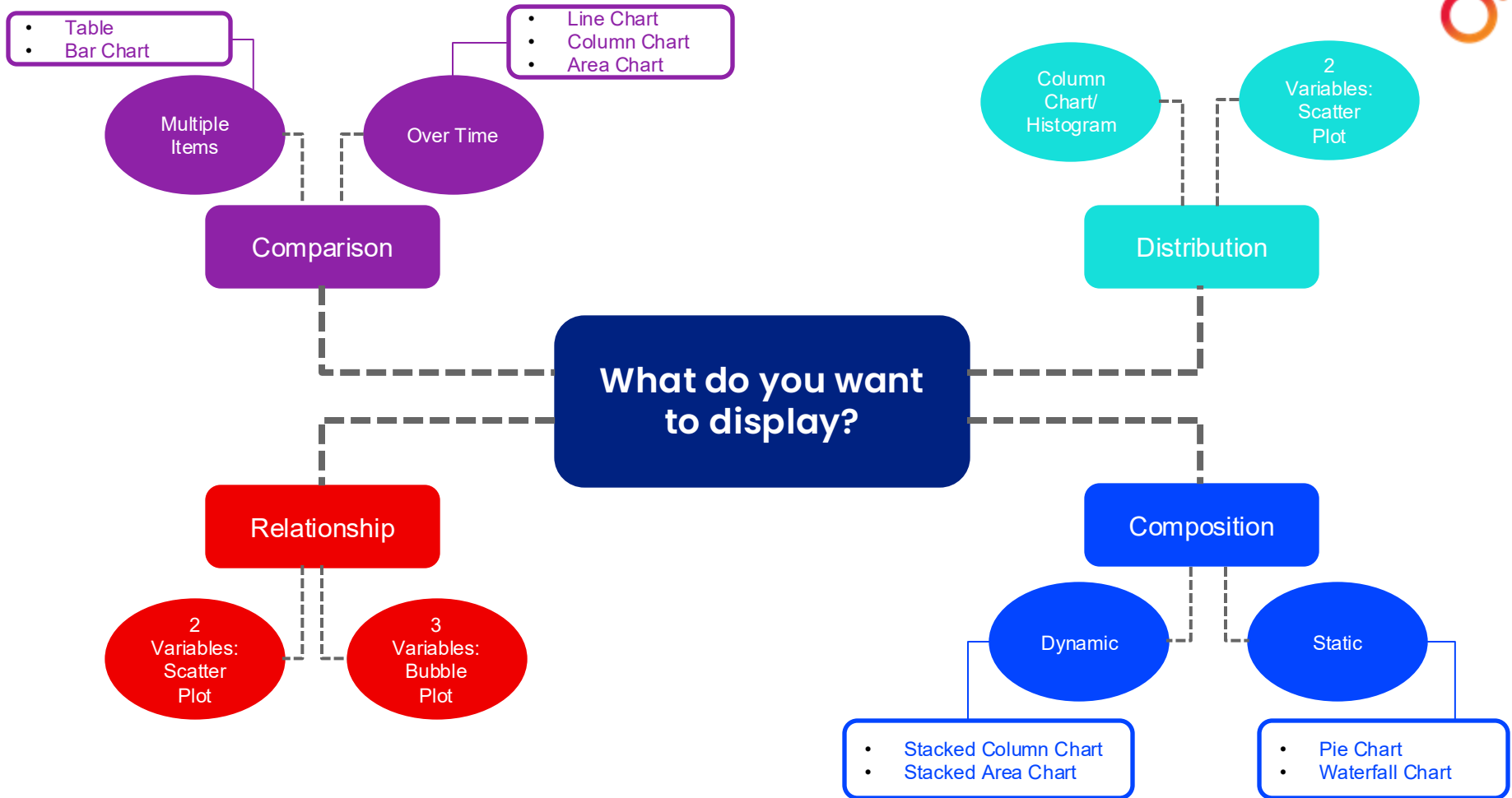
Introduction and Recap



Creating Reports



**Think about the message you
want to share with the audience**



Motivation for Conditional Formatting



Provide additional context

**Customize cell color, data bars,
KPI icons**



Highlight the data

Enable users with quick insights



Slicers vs Filters

Slicers



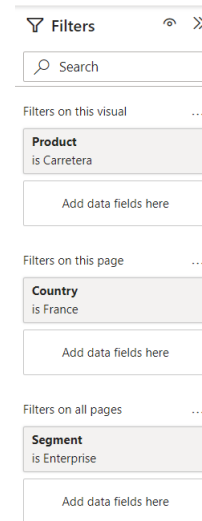
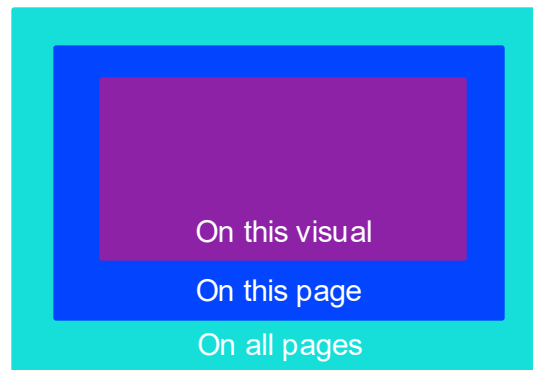
Quick access

Focused reports

Simplify filtered view

Filters

You, the report designer, define filters!





Slicers vs. Filters – When to Use What?

Slicers

- ✓ Display common filters on the report page
- ✓ Focused view
- ✓ Simplified filtered state

Filters

- ✓ Complex filtering requirements
- ✓ Limited space on the report page
- ✓ Lock/hide aspects of the Filter pane



Report Page Configuration

Change page name/Instruct Power BI to use the page as a tooltip or Q&A

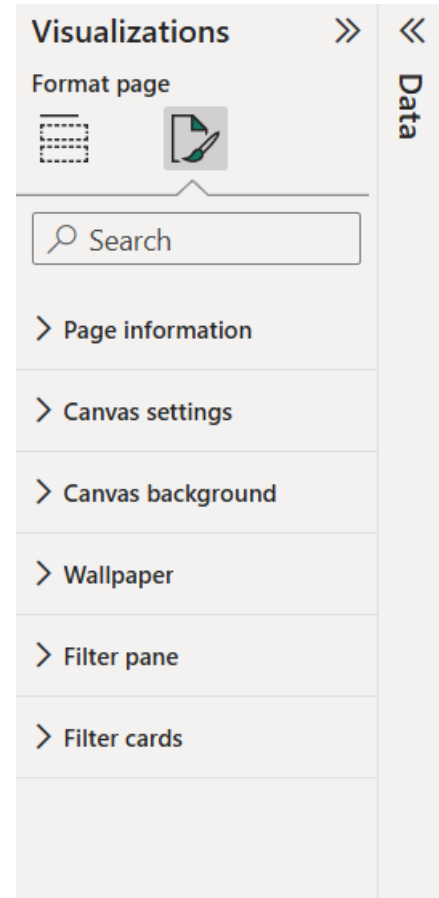
Change the default size of the page

Change the color, transparency, set image as a background

Similar to background, but it stretches outside of the page

Customize the look of the entire Filter pane

Customize the look of Filter cards





Choosing the Right Tool – Paginated Report



Data represented in tables



Sales invoice to be printed or exported to PDF



All data in a single report element



Development in Power BI Report Builder



Paginated Report vs “Classic” Power BI

Paginated report

Development in Power BI Report Builder

Print or export all the data

No underlying data model

No shared data sources and shared datasets

Power BI

Development in Power BI Desktop

Print or export only the visible data

Has its own data model

Possibility to share data sources and datasets between the reports



Create Visual Calculations



Excel-like experience when using DAX!

Reference value from
cells

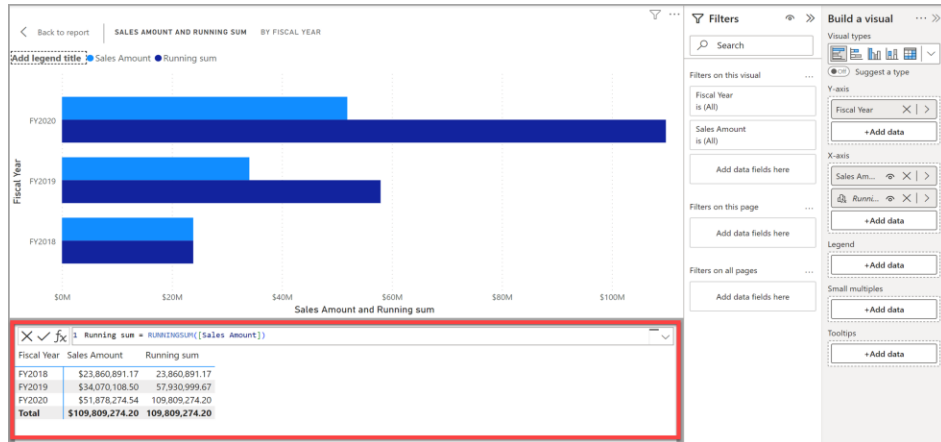




How Visual Calculations work?

- DAX calculation defined and executed directly on the visual level
- Stored on the visual, not in the model!
- Can refer only to visual fields
- Operate on aggregated instead of the detailed data

Use templates





What Is a Copilot?

**Copilot is Microsoft's implementation of LLMs
(Large Language Models)**



Enable Copilot



Microsoft Fabric

Search

Admin portal

- Tenant settings **New**
- Usage metrics
- Users
- Premium Per User
- Audit logs
- Domains **New**
- Workloads
- Tags (preview) **New**
- Capacity settings
- Refresh summary
- Embed Codes
- Organizational visuals
- Azure connections
- Workspaces
- Custom branding
- Protection metrics
- Fabric identities
- Featured content
- Help + support

There are new or updated tenant settings. Expand to review the changes.

Copilot and Azure OpenAI Service

Users can use **Copilot** and other features powered by Azure OpenAI
Enabled for the entire organization

When this setting is enabled, users can access the features powered by Azure OpenAI, including **Copilot**. This setting can be managed at both the tenant and the capacity levels. [Learn More](#)

For customers in the EU Data Boundary, this setting adheres to Microsoft Fabric's EU Data Boundary commitments. [Learn More](#)

By enabling this setting, you agree to the [Privacy Terms](#).

☒ Enabled

Note: Copilot in Fabric is now generally available, starting with the Microsoft Power BI experience. The Copilot in Fabric experiences for Data Factory, Data Engineering, Data Science, Data Warehouse, and Real-Time Intelligence are in preview.

Note: If Azure OpenAI is not available in your geographic region, your data may need to be processed outside your capacity's geographic region, compliance boundary, or national cloud instance. To allow data to be processed outside your capacity's geographic region, turn on the related setting, "Data sent to Azure OpenAI can be processed outside your capacity's geographic region, compliance boundary, or national cloud instance".

Apply to:

☒ The entire organization

☐ Specific security groups

☐ Except specific security groups

Delegate setting to other admins

Select the admins who can view and change this setting, including any security group selections you've made.

☐ Capacity admins can enable/disable

Apply Cancel

Prepare Semantic Model for Copilot



Prepare Semantic Model for Copilot



Object	Recommendation	Consideration	Example
Table	Define clear relationships	Ensure proper relationships are established, including relationship cardinality (1:1, 1:M, M:M)	The Sales table is connected to the Customer table on the CustomerKey column
Measure	Standardized calculation logic	Ensure standardized and clear calculation logic	The Total Sales measure calculated as the SUM of the SalesAmount column from the Sales table
Measure	Naming conventions	Ensure the measure name clearly reflects the calculation	Use "Average_Tax_Amount" instead of "AvgTaxAmt"
Fact table	Create a fact table in accordance with the Star schema design	Fact tables should contain measurable data in addition to dimension keys	Orders, Sales, Purchases
Dimension table	Create a dimension table in accordance with the Star schema design	Dimension tables should contain descriptive attributes relevant to the object of interest	Customer, Product, Date
Column	Use clear names	Ensure column names are unambiguous and self-explanatory	Use Customer_Name instead of CustomerKey



DEMO

- Configure report page and visuals
- Apply custom report theme
- Configure conditional formatting
- Implement slicers and filters
- Create visual calculations
- Use Copilot to create report pages





Q&A

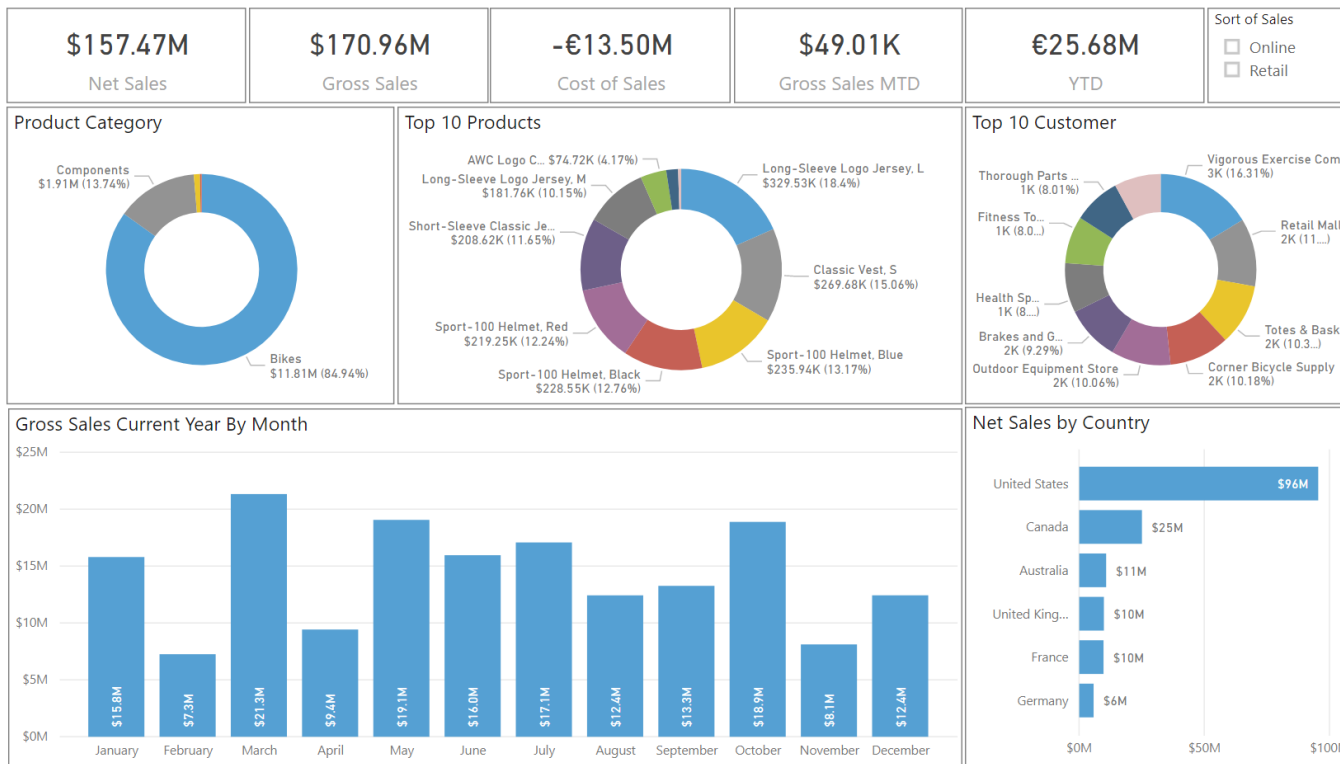




Enhance Reports for Usability and Storytelling



Design an Appealing Report Layout

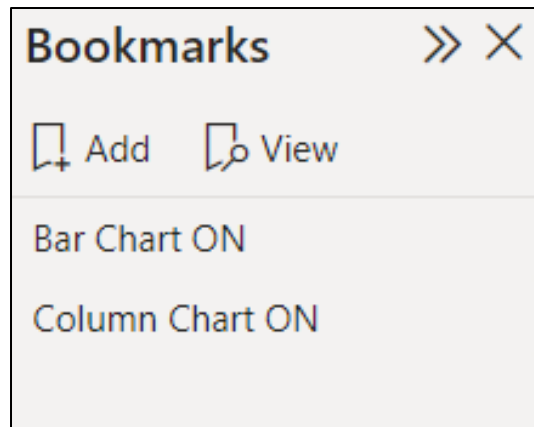




Bookmarks

Capture the current state of the report page

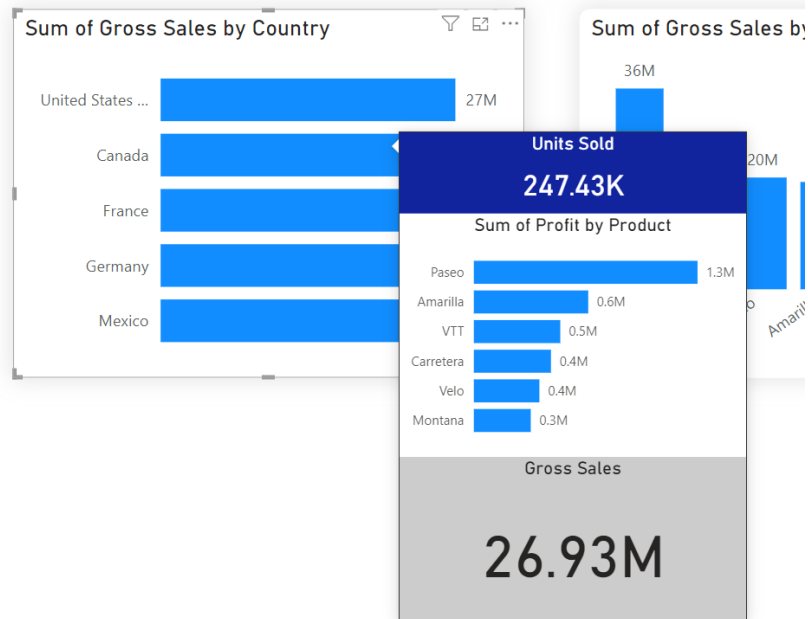
- ✓ Current page
- ✓ Filters
- ✓ Slicers
- ✓ Sort order
- ✓ Drill location





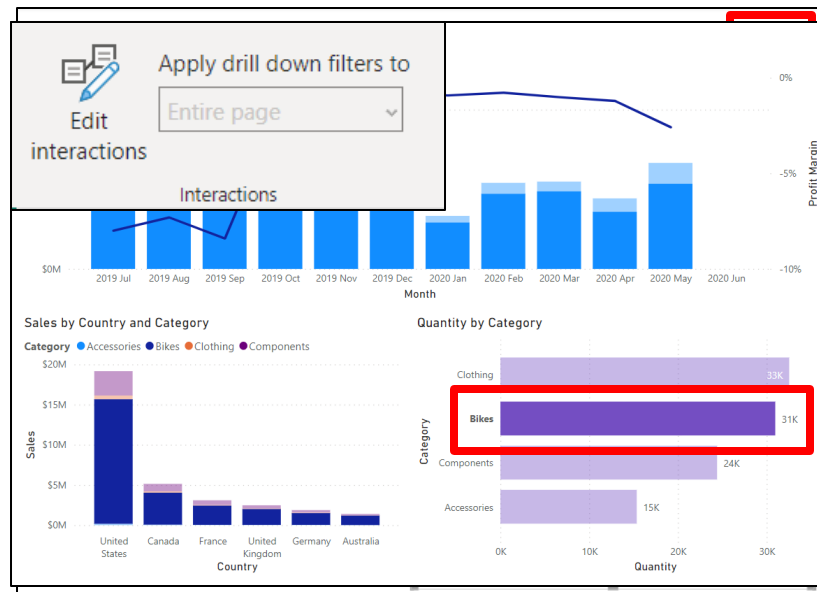
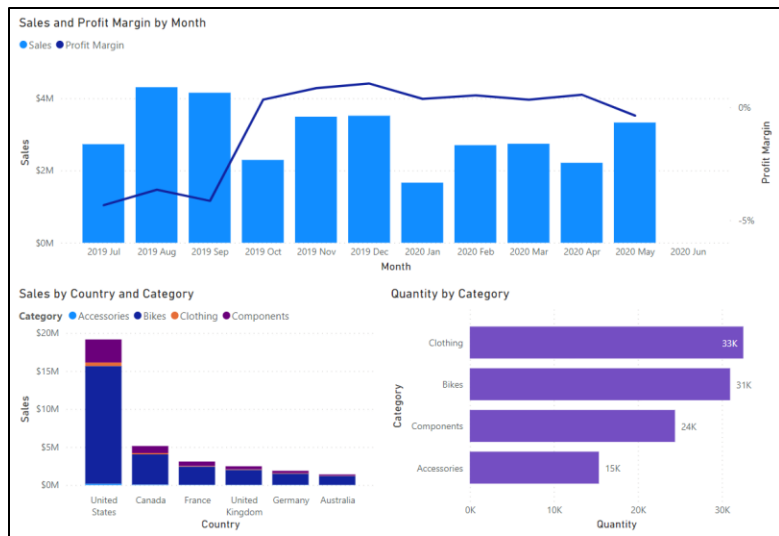
Custom Tooltips Scenarios

- ✓ Visually rich tooltips
- ✓ Include visuals, images, and other report page elements





Cross-highlight Between Visuals





Report Navigation with Buttons

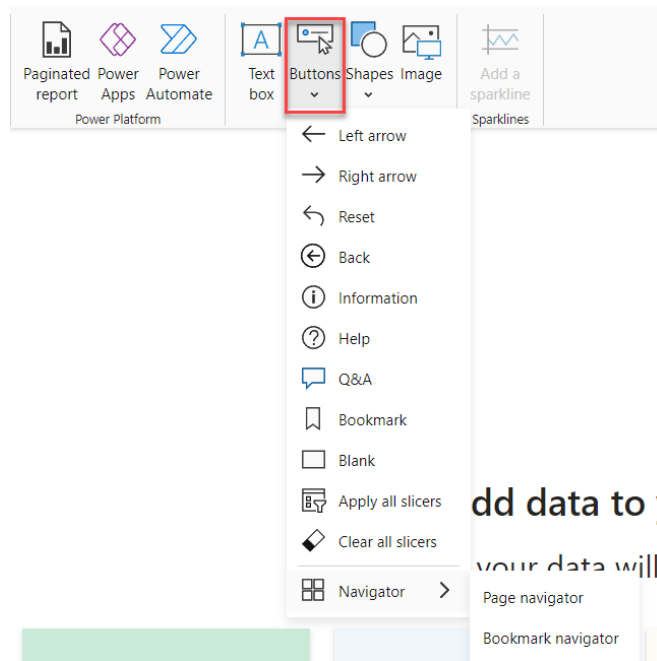
✓ **Back**

✓ **Bookmark**

✓ **Reset**

✓ **Navigator -> Page**

navigator/Bookmark navigator





Sync Slicers

- ✓ Slicer selections on page A will propagate to Page B, even though the slicer is not visible on Page B!
- ✓ Independently configure and format the synced slicer on Page B

Sync slicers

Add and sync with all pages, or select specific pages:

Page name		
Page A	☐	☒
Page B	☐	☐

Advanced options

Enter a group name to sync selection to any other visuals with that group name

☐ Sync field changes to other slicers

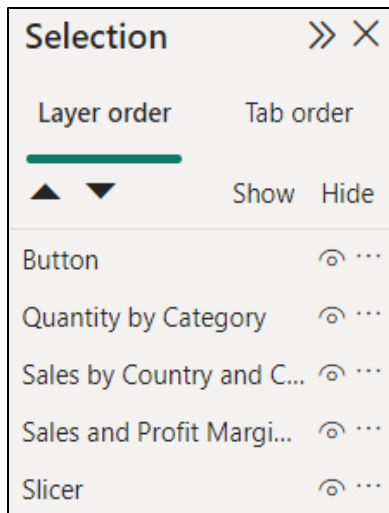
☐ Sync filter changes to other slicers



Selection Pane

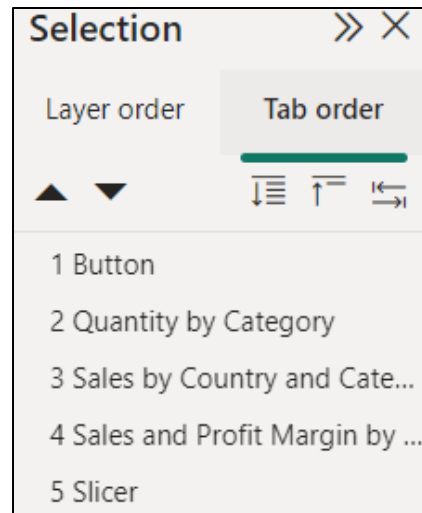
Layer order

Defines the order of visuals

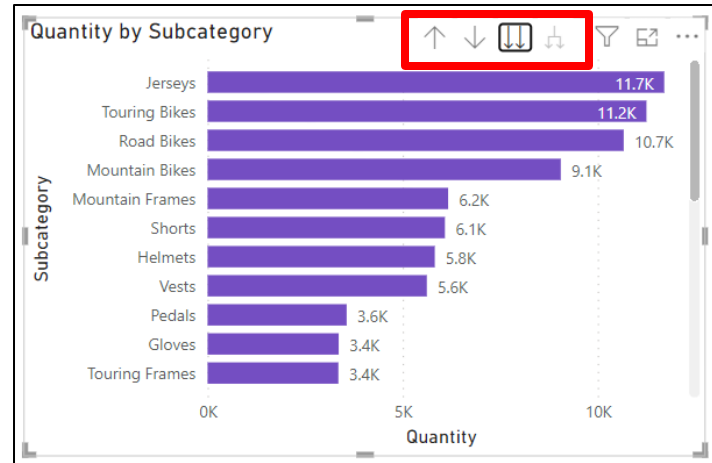
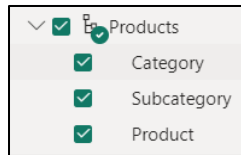
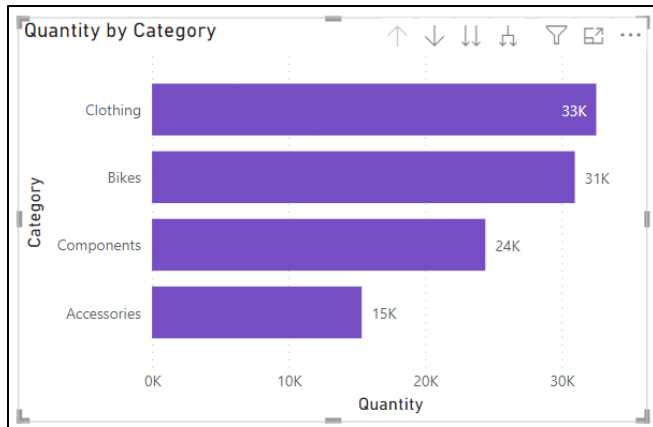


Tab order

Controls keyboard user navigation

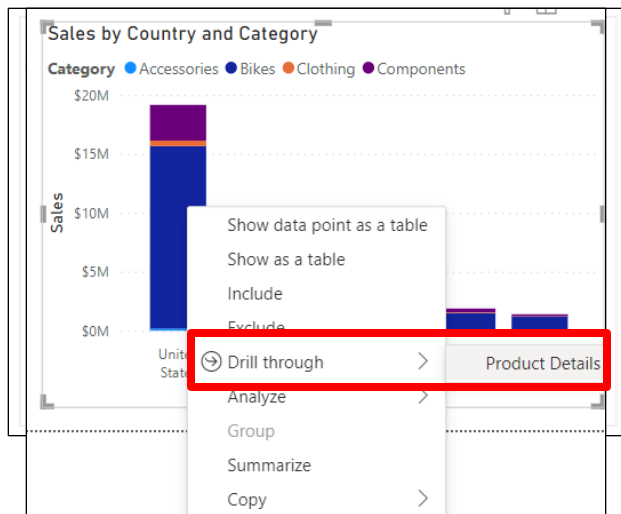


Drill down Through Visual Using a Hierarchy





Drill through



Bikes				
Subcategory	Color	Quantity	Sales	Profit Margin
Mountain Bikes	Black	2,501	\$2,484,496	8.69%
Mountain Bikes	Silver	3,384	\$2,603,065	7.56%
Road Bikes	Black	3,040	\$2,389,419	-6.13%
Road Bikes	Red	205	\$300,532	-6.07%
Road Bikes	Yellow	3,938	\$3,380,758	-6.60%
Touring Bikes	Blue	2,988	\$2,627,332	-5.65%
Touring Bikes	Yellow	2,065	\$1,696,351	-16.63%
Total		18,121	\$15,481,954	-2.62%

Drill through

Cross-report ☐ Off

Keep all filters ☐ On

Category is Bikes

Country is United States

Year is FY2020



“Can I export this report to...?”





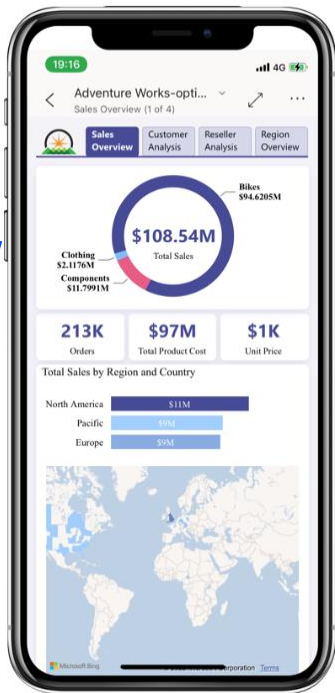
“Can I export this report to...?”



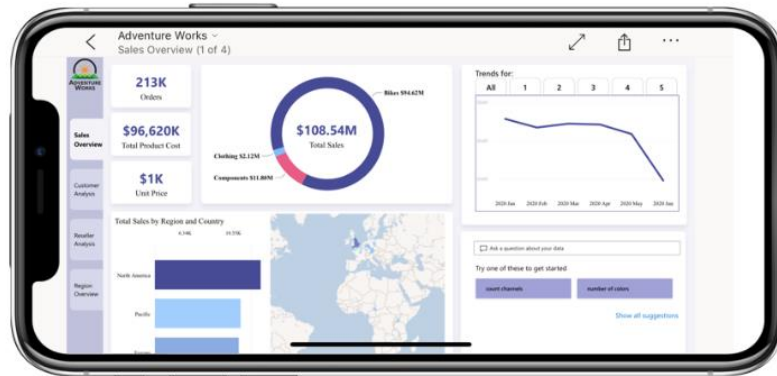
- Exports data to .csv
- Filters are preserved
- Sensitivity labels respected
- Max number of exported rows to **csv** = 30.000
- Max number of exported rows to **xl/sx** = 150.000

- Each page of the report is an individual pdf page
- Sensitivity labels respected
- **Current values** exports filtered data
- **Default values** exports the original state
- Not all visuals are supported

Creating Mobile-optimized Reports



Mobile-optimized
view



Regular, non-optimized view



Enable Personalized Visuals



**Report consumers can
customize the visual
according to their needs**



Enabling Personalized Visuals

- ✓ **Sharing** – “personalized” version will not overwrite the original
- ✓ **Reset changes** – Revert the whole report or only a specific visual



Design for Accessibility



Why Accessibility Matters?



No visual impairments



Green-blind (Deuteranopia)



Accessibility Features in Power BI

Built-in

- ✓ Keyboard navigation
- ✓ Screen reader
- ✓ High contrast color
- ✓ Focus mode
- ✓ Show data table

Configurable

- ✓ Alt text
- ✓ Tab order
- ✓ Labels & titles
- ✓ Markers
- ✓ Themes, contrast, colors



Accessibility Checklist



- ✓ Ensure color contrast is at least 4.5 : 1
- ✓ Avoid using ONLY colors for transmitting the information
- ✓ Use clear textual descriptions
- ✓ Add Alt text to all non-decorative visuals
- ✓ Avoid tooltips for conveying the important information



Configure Automatic Page Refresh



**Scheduled refresh of the
specific report page!**



Auto Page Refresh

Works only with DirectQuery

Fixed Interval

- ✓ Set desired interval (1 sec to X days)
- ✓ All visuals refreshed



Change Detection

- ✓ Specify a measure to be monitored
- ✓ ONE SINGLE query





Auto Page Refresh Considerations

- ✓ Think about expected data change frequency
- ✓ Workload created by queries
 - 20 concurrent users x 3 visuals = 60 queries every 2 seconds!
- ✓ Change detection only with Premium
- ✓ Change detection only on 1 measure per dataset
- ✓ Max 10 data models with Change detection in tenant



DEMO

- Create and configure bookmarks
- Create custom tooltips
- Edit interactions between the visuals
- Configure sync slicers
- Implement Drill through feature
- Configure Auto page refresh
- Enable personalized visuals





Q&A





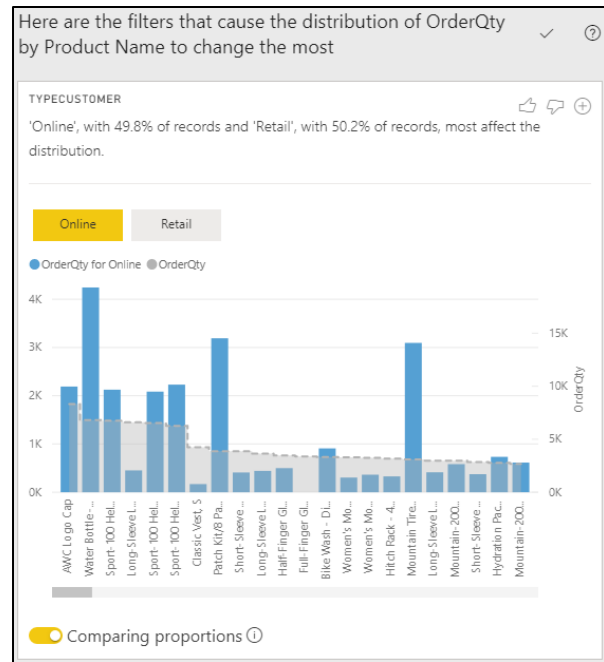
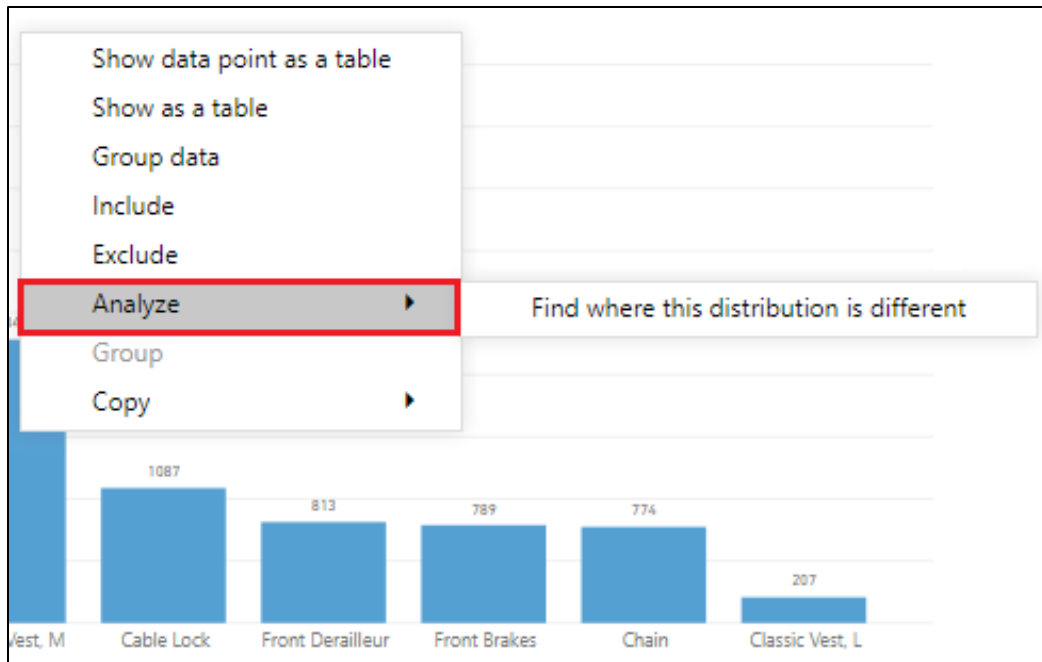
Break



Identify Patterns and Trends in Data



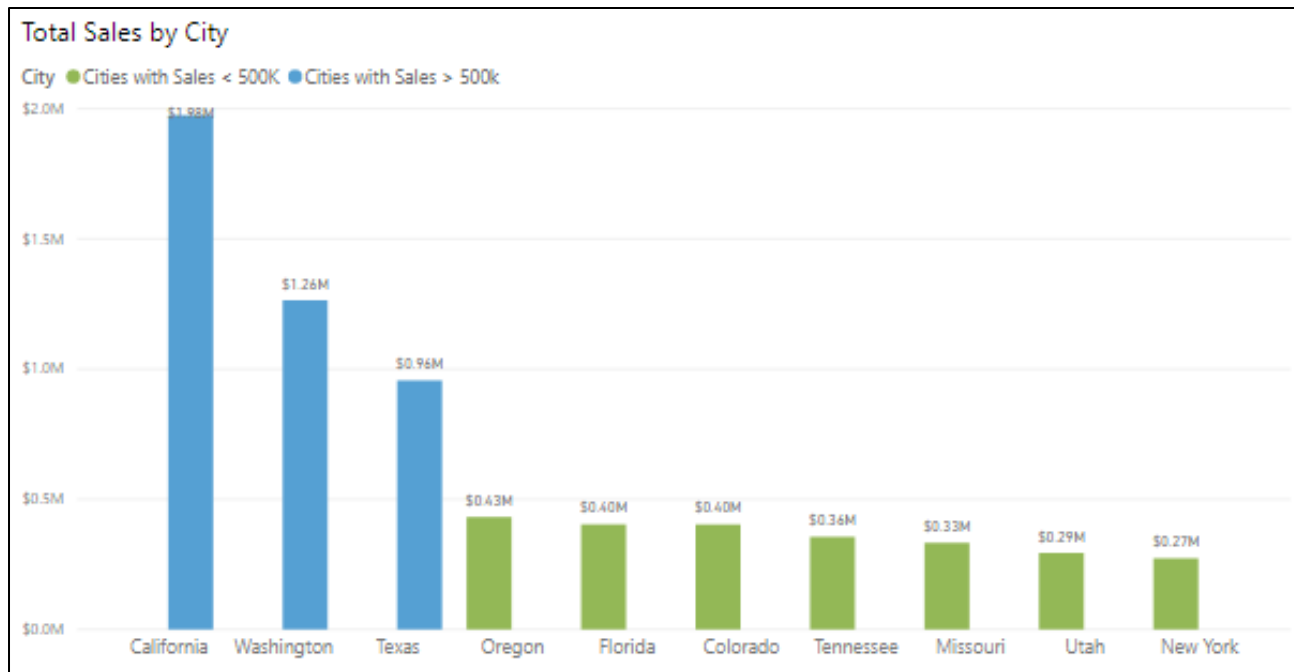
Analyze feature





Grouping and Binning

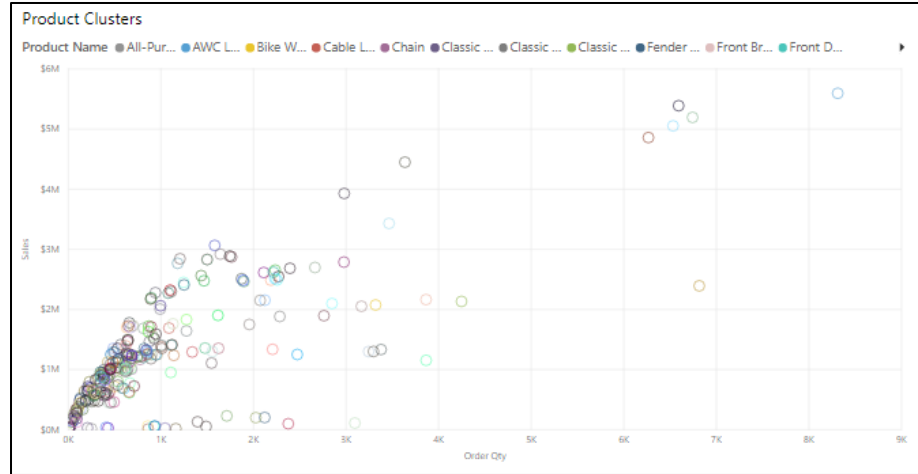
Organize data into categories or intervals to better analyze the data





Clustering with Scatter chart visual

- Automatic data grouping
- Enhance pattern recognition and insights
- Simplify data understanding and analysis



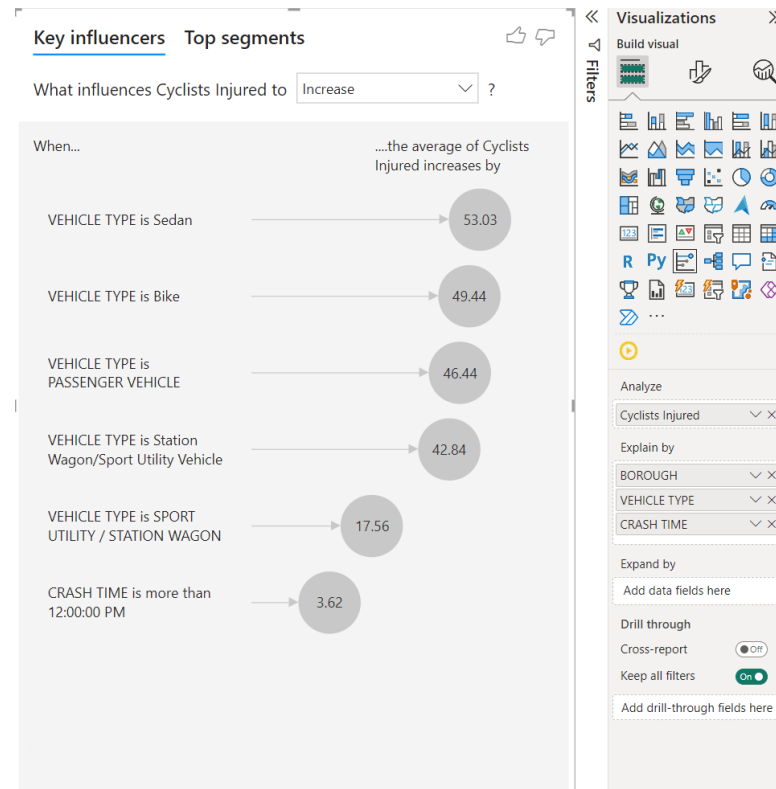


Use AI Visuals



Key influencers Visual

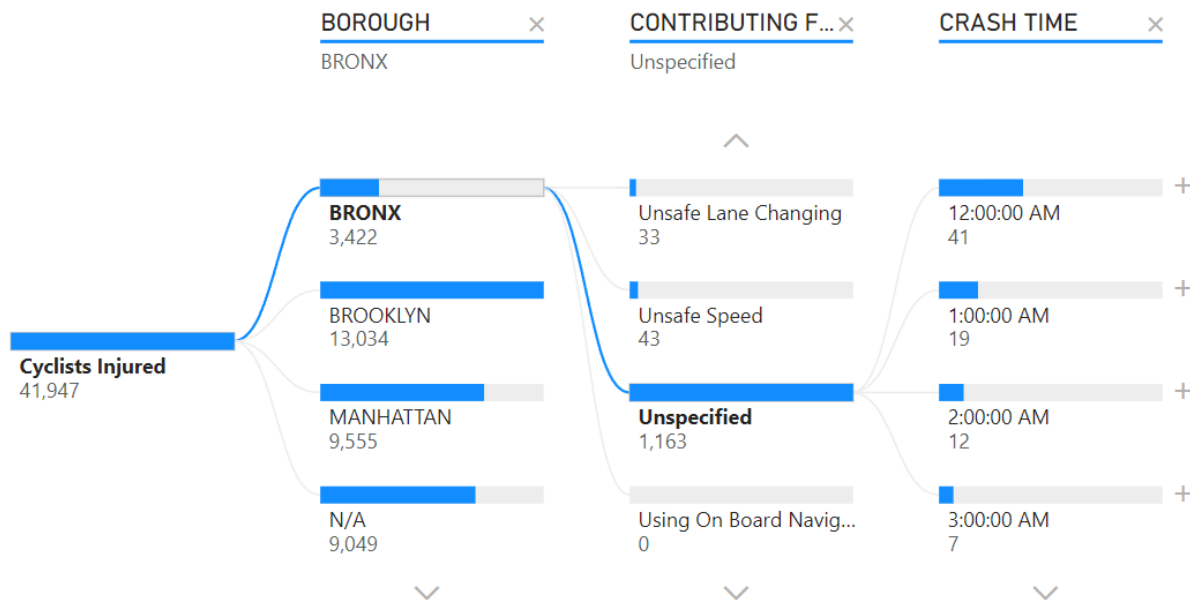
Enables identifying factors behind variations in a chosen target metric





Decomposition Tree Visual

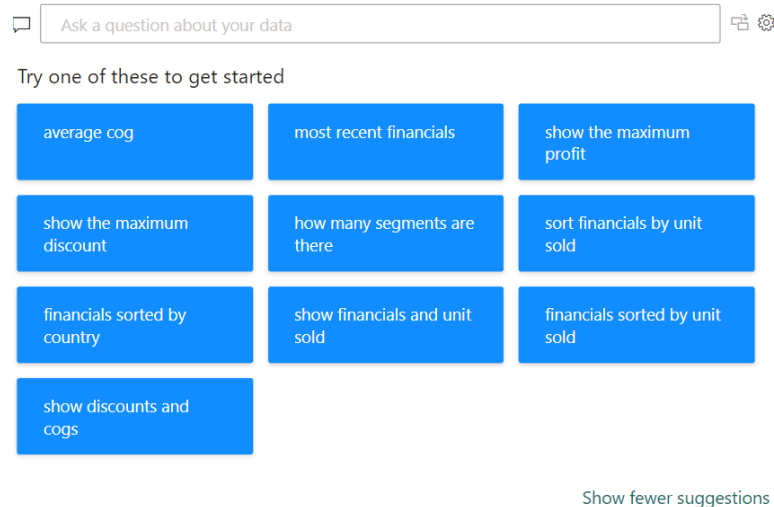
Visualize data hierarchies for in-depth exploration and pattern discovery





Q&A Visual

- ✓ Question bar
- ✓ List of suggested questions
- ✓ Convert Q&A visual into a regular visual





Implement Reference Lines, Error Bars, and Forecasting



Implement Reference Lines

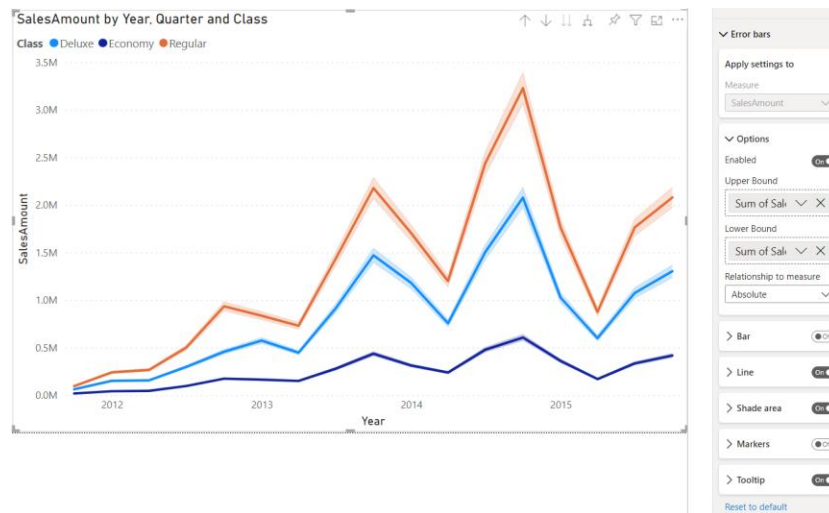
- ✓ Quickly understand data points compared to a reference
- ✓ Available options depend on the column data type





Implement Error Bars

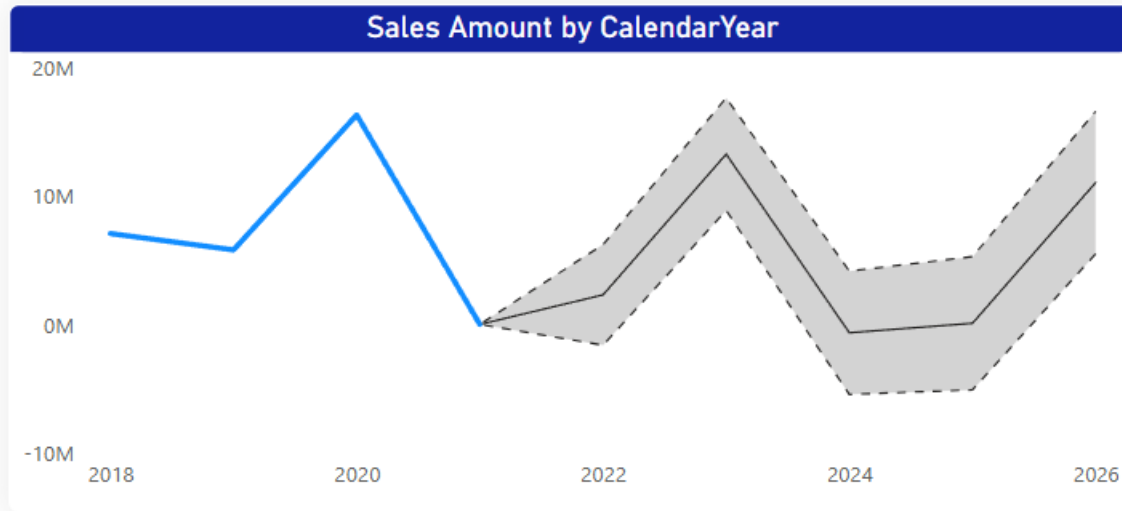
- ✓ Visualize the *uncertainty* in the data
- ✓ Uncertainty = displayed value contextualized by the range of upper and lower bounds





Implement Forecasting

- ✓ Prediction of future trends based on the current data



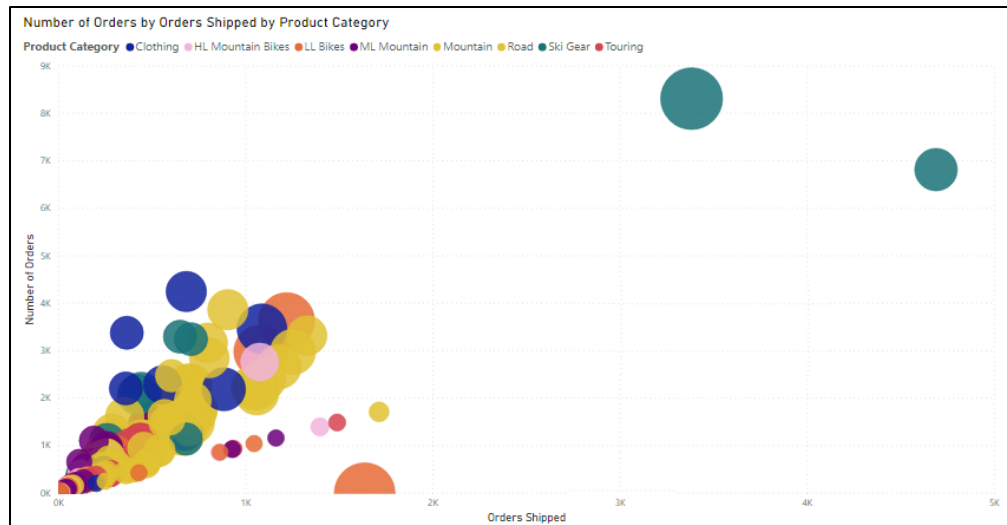


Detect Outliers and Anomalies



Identify Outliers

- Unexpected deviations in data
- Scatter charts can detect outliers
- Use DAX to define and analyze outliers dynamically





DEMO

- Use Analyze feature
- Implement grouping and binning
- Implement AI visuals
- Implement reference lines, error bars and forecasting





Q&A





Create and Manage Workspace and Assets



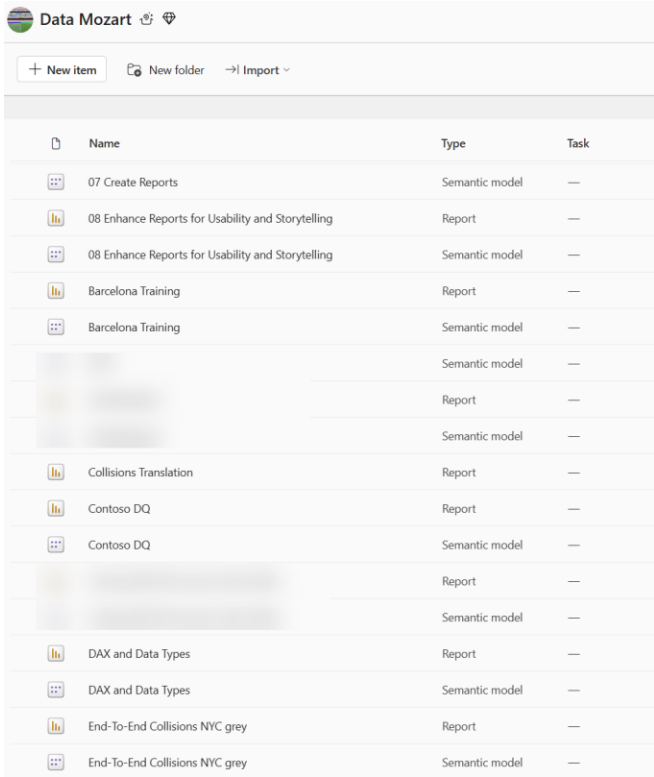
**Workspace is a
container for all
Power BI items**





Workspaces in a Nutshell

- Share and present reports and dashboards
- Ensure security by controlling access
- Enforce collaboration -> Different teams can work together in the workspace



The screenshot shows the Data Mozart workspace interface. At the top, there's a header with the Data Mozart logo and a search icon. Below the header, there's a toolbar with buttons for '+ New item', 'New folder', and 'Import'. The main area displays a table of workspace items.

	Name	Type	Task
	07 Create Reports	Semantic model	—
	08 Enhance Reports for Usability and Storytelling	Report	—
	08 Enhance Reports for Usability and Storytelling	Semantic model	—
	Barcelona Training	Report	—
	Barcelona Training	Semantic model	—
		Semantic model	—
		Report	—
		Semantic model	—
	Collisions Translation	Report	—
	Contoso DQ	Report	—
	Contoso DQ	Semantic model	—
		Report	—
		Semantic model	—
	DAX and Data Types	Report	—
	DAX and Data Types	Semantic model	—
	End-To-End Collisions NYC grey	Report	—
	End-To-End Collisions NYC grey	Semantic model	—

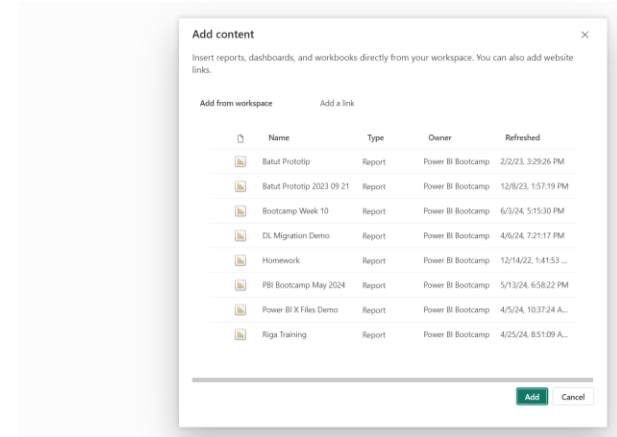
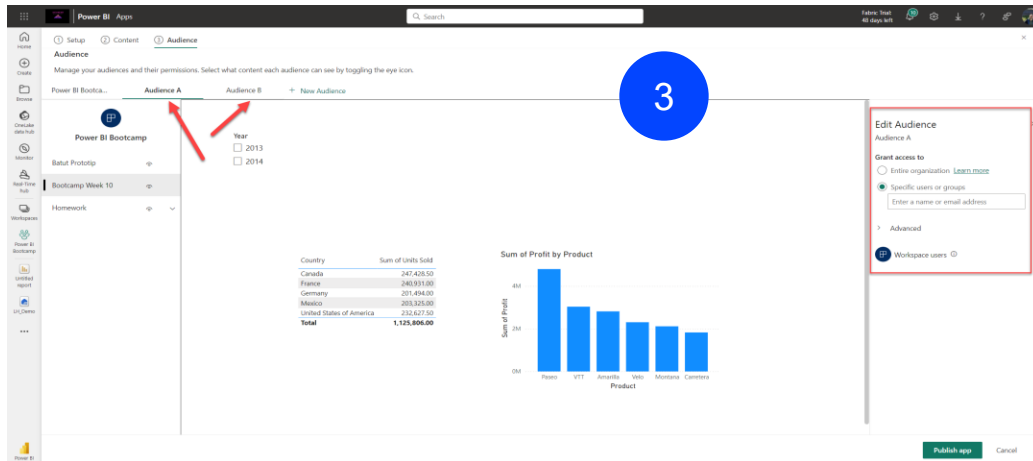
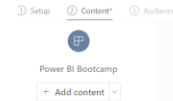
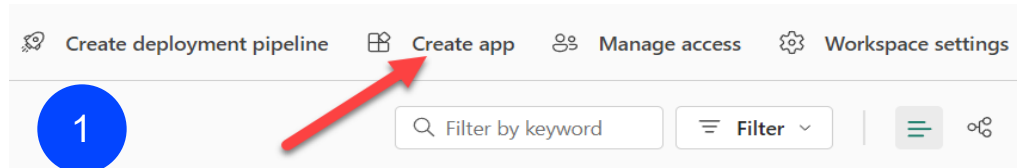


Configure a Workspace App



Create Apps

Use apps for content distribution!





Create Dashboards



What is a Power BI Dashboard?

- Single page item
- Visuals pinned from reports
- Tiles are static
- Refreshes with semantic model
- DirectQuery every 15 min



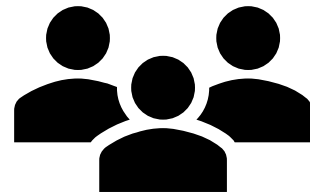


Understand Distribution Methods



Distribute Dashboards and Reports

- **Share reports and dashboards from a workspace**
- **Present live in PowerPoint**
- **Print reports**
- **Distribute in the Power BI app**
- **Embed reports – custom apps or Publish to Web**





Configure Subscriptions and Alerts



Subscribe to Power BI Report

Subscribe to report

609
aths

1M
Injuries

Collisions by Year/Month

Top 5 collision causes

Distraction 346K

ht-of-Way 104K

oo Closely 93K

g Unsafely 68K

Vehicular 57K

263.56K

212K

Sunday

Subscriptions

Keep track of your data by subscribing to this report.

Manage all

Main

Unsaved changes

Subscription name *

Main

Recipients *

Nikola Ilic

Attach full report

None

Scheduled date and time

Start date *

9/6/2024

End date

Select a date

Repeat *

Daily

Scheduled time *

3 30 PM

Time zone *

(UTC+01:00) Brussels, Copenhagen, Ma

Emails will be sent daily at 03:30 PM (UTC+01:00) Brussels, Copenhagen, Madrid, Paris starting Friday, September 6, 2024.

More options

Save New subscription



Configure Alerts

➤ Dashboard tiles

- *My workspace*
- Shared with you in a Premium capacity
- Any workspace you can access if you have a Pro or PPU license

➤ Can be set on gauges, KPIs, and cards

➤ Only work with numeric data types



Promote and Certify Power BI Content



Promote and Certify Power BI Items

Promotion

- Highlight items valuable and worthwhile for others to use
- Encourages the collaborative use and spread of content within an organization
- Any item owner and users who have write permissions can promote the item

Certification

- Item meets organizational quality standards and can be regarded as reliable and ready for use across the organization
- Only authorized reviewers (defined by the administrator) can certify items



Understand Gateways

Bridge between the Power BI Service and on-premises data sources





Data Gateway Types

On-prem

- ✓ Multiple users -> Multiple on-prem data sources
- ✓ Single installation
- ✓ Suitable for enterprises

On-prem (Personal)

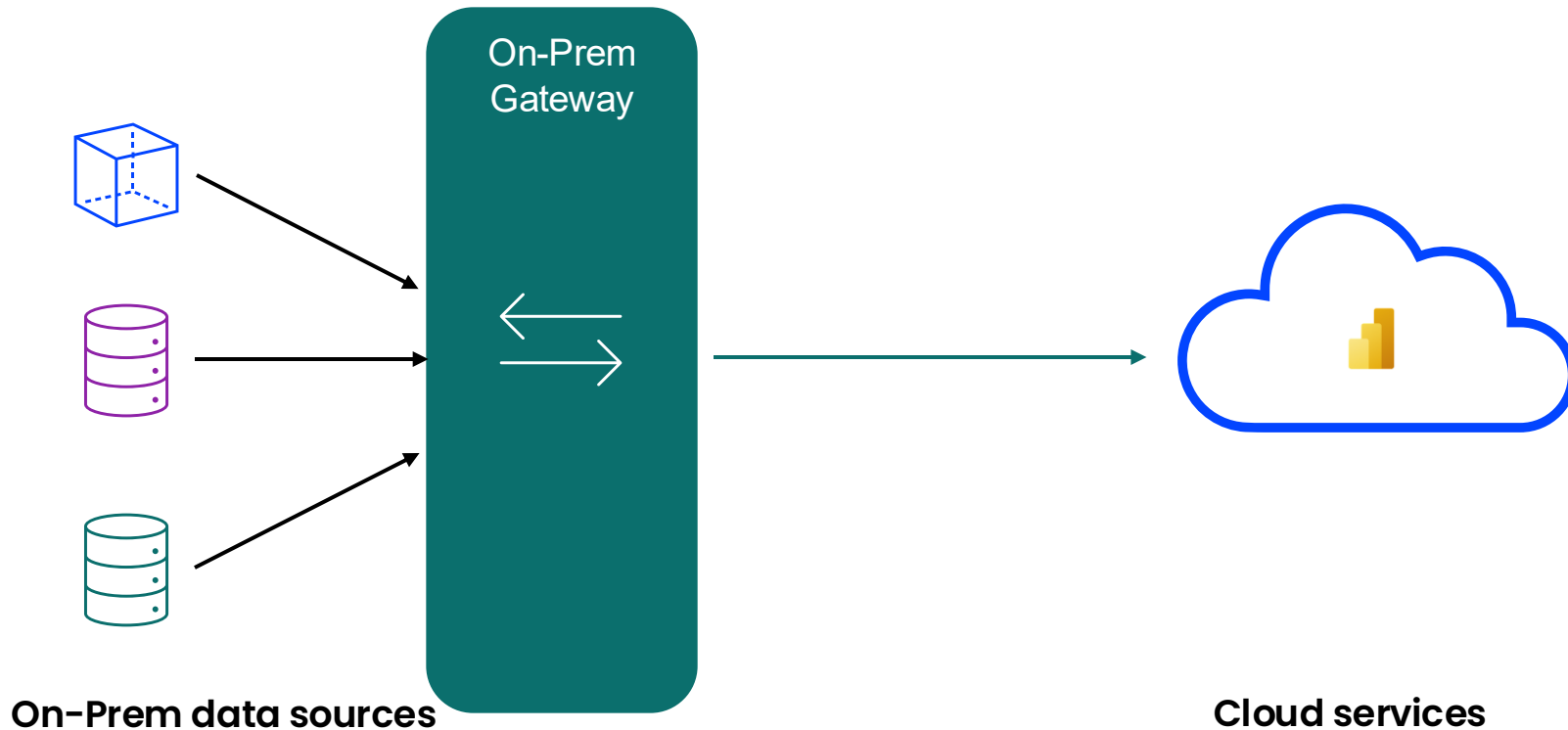
- ✓ 1 user -> Multiple on-prem data sources
- ✓ Works only with Power BI
- ✓ Only person creating reports

VNet

- ✓ Multiple users->Multiple data sources
- ✓ Data sources part of the VNet
- ✓ No installation needed

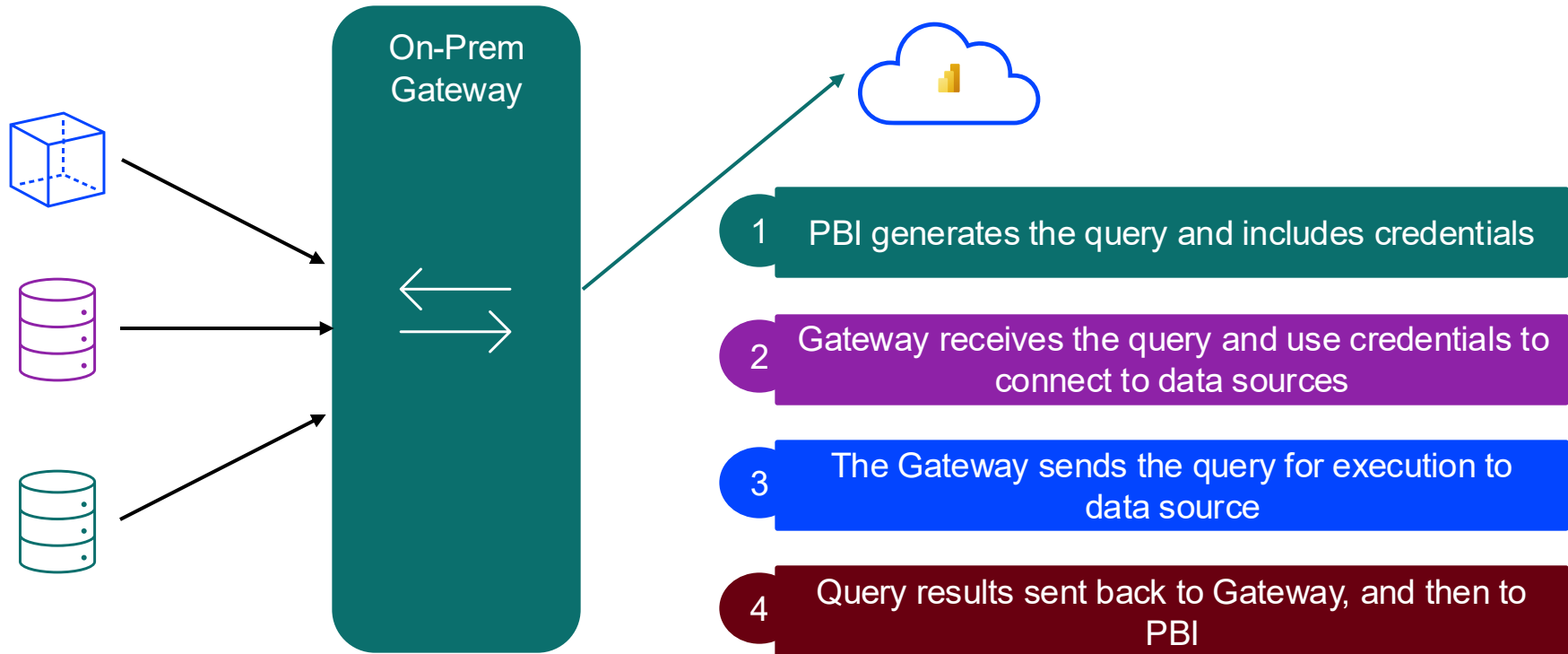


How Does the Gateway Work?





How Does the Gateway Work?





Configure Scheduled Refresh



**8x per day with
shared capacity**

**48x per day with
Premium capacity**





Scheduled Refresh Issues

Disabled

- ✓ 4 consecutive failures
- ✓ Configuration update ->
Invalid/expired credentials

Paused

- ✓ 2 months of inactivity



Secure and Govern Power BI Items

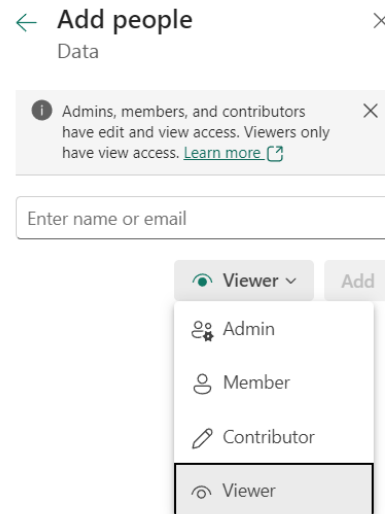
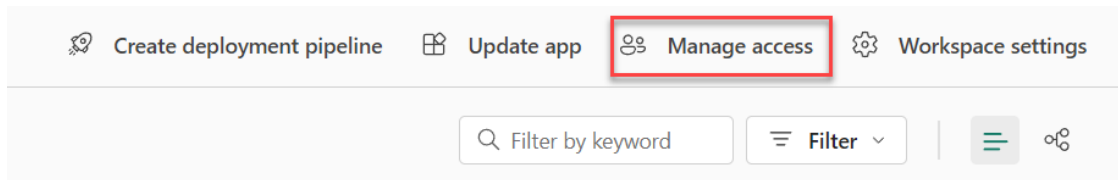


Assign Workspace Roles



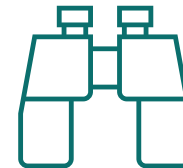
Assign Workspace Roles

Use workspaces for collaboration, not content sharing!





Viewer Role



- ✓ View and interact with the item
- ✓ Read data from the dataflows
- ✓ Create report subscriptions



Contributor Role

Viewer +...

- ✓ Publish, create, edit, and delete content
- ✓ Create reports in another workspace
- ✓ Schedule data refresh
- ✓ Modify gateway connection settings
- ✓ Use the Analyze in Excel feature





Member Role

Contributor +...



- ✓ Add members or others with lower permissions
- ✓ Publish/unpublish the app
- ✓ Share items in the app
- ✓ Allow others to reshare items
- ✓ Manage semantic model permissions



Admin Role

Member +...

- ✓ Update and delete the workspace
- ✓ Add or remove any user in a workspace role
- ✓ Allow contributors to update the app
- ✓ Manage subscriptions created by others





Configure Access to Semantic Models



Understanding the Build Permission

- ✓ New content based on the shared semantic model
- ✓ Export underlying data
- ✓ Use the Analyze in Excel feature
- ✓ Access data via XMLA endpoint





Implement Row-level Security Roles



Row-level security limits access to specific attributes (country, product...)





Static vs Dynamic RLS

Static

- ✓ Basic scenarios
- ✓ Rules maintained in the PBIX file
- ✓ Easy and straightforward setup



- ✓ Maintenance, lot of manual work, not reusable

Dynamic

- ✓ Control access on a granular level
- ✓ Access control via data model
- ✓ Reusable and less maintenance



- ✓ Adds to model complexity

USERNAME() and USERPRINCIPALNAME()



RLS Considerations

- ✓ One user in multiple roles – filters work in additive way
- ✓ Performance impact – enforce RLS filters on dimension tables
- ✓ Assign Entra ID groups instead of individual accounts
- ✓ Configurable for Import, DirectQuery, and Direct Lake models, **NOT** for Live Connect



Configure Row-level Group Membership



Adding Member Groups to RLS Roles

- ✓ Distribution group
- ✓ Mail-enabled group
- ✓ Security group

Users/groups with Admin, Member, or Contributor role in the workspace will see ALL data, regardless of RLS rules!



Implement Sensitivity Labels



Sensitivity Labels

- Must be defined in Microsoft Purview Information Protection
- Must be enabled in the Admin Portal
- Can be applied both in Power BI Service and Desktop
- Apply to individual report/dashboard Settings

⚡ Allow users to apply **sensitivity** labels for content

Unapplied changes

With this setting enabled, Microsoft Purview Information Protection **sensitivity** labels published to users by your organization can be applied. All [prerequisite steps](#) must be completed before enabling this setting.

Note: **Sensitivity** label settings, such as encryption and content marking for files and emails, are not applied to content. [Learn More](#)

Visit the [Microsoft Purview compliance portal](#) to view **sensitivity** label settings for your organization.

Note: **Sensitivity** labels and protection are only applied to files exported to Excel, PowerPoint, or PDF files, that are controlled by "Export to Excel" and "Export reports as PowerPoint presentation or PDF documents" settings. All other export and sharing options do not support the application of **sensitivity** labels and protection.

☒ Enabled

ⓘ The setting below determines which users in the organization can apply and change sensitivity labels. All other users in the organization can only view the labels.

Sensitivity labels don't affect access to the content!



DEMO

- Assign workspace roles
- Configure the app
- Create a dashboard
- Configure semantic model scheduled refresh
- Configure membership for row-level security





Q&A



It's your turn...



Hands-on Lab Part 2



[Lab 3 Instructions](#)

[Lab 4 Instructions](#)

[Lab 5 Instructions](#)

[Lab 6 Instructions](#)





Practical Exam Tips, Quiz, and Closing



Practical Exam Tips

- ✓ Make sure to prepare the room (for online takers)
- ✓ You can use Microsoft Learn during the exam 😊...
- ✓ ...but, don't let this waste too much of your time!
- ✓ Take time to understand case studies and explore the exhibits/datasets before answering (there is no way back)

Quiz





Join the Quiz!



<https://app.sli.do/event/nR8vJfCPY4PJDkLVzkRu4Y>



Group Discussion

- Which benefits do you expect from passing the PL-300 exam?



Summary



Takeaways

- Check the official Microsoft PL-300 study guide:
 - [Official study guide](#)
- [PL-300 Certification page with practice assessment](#)
 - Take the practice assessment
 - Experience demo
- Microsoft Power BI courses and books @ O'Reilly platform
 - [Self-paced PL-300 course](#)
- Take the next step with Microsoft Fabric
 - DP-600 – Fabric Analytics Engineer (includes Power BI)
 - DP-700 – Fabric Data Engineer (NO Power BI!)





Q&A



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